Universidad de O'Higgins
School of Engineering
Electrical Civil Engineering

Implementation and Validation of a Scheme
Geographically distributed co-simulation for real-time electrical system analysis based on OPAL-RT and RTI Connext DDS

Sebastián Escobar Rojas
Thesis Advisor: Dr. Claudio Burgos

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Dedication

To Rosita Moreno Moreno

Your memory accompanies me in every achievement,

And this memory, like so many other things in my life, silently bears your name.

Acknowledgments

I want to express my sincere gratitude to Professor Claudio Burgos, who was the first to offer me a helping hand when I was looking for an opportunity to complete my thesis. His willingness, guidance, and confidence were fundamental throughout this entire process.

I am deeply grateful to my parents for their unconditional love and support, to my grandmother, whose memory accompanied me while I worked through this stage and whose passing during my university years left an indelible mark, to my sister, an example of effort and determination, who inspires me to keep going, to my nephews, because their existence gives meaning and joy to my life, to my partner, for their constant support, their tireless companionship and for walking beside me throughout this process, and to my friends, because without you university would have been just studying.

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1 Table of Contents

Summary

This research presents a real-time distributed co-simulation architecture for electrical systems, integrating MATLAB/Simulink (a block diagram-based programming and graphical simulation environment that allows modeling, simulation, and analysis of dynamic systems), OPAL-RT OP4200 (a real-time simulation platform used for Hardware-in-the-Loop (a simulation technique in which a real physical system (hardware) is connected to a real-time computational model)). OPAL-RT's digital simulators (OP4200) allow Simulink models to be run on dedicated hardware. The architecture also incorporates RTI Connext DDS middleware (a communication middleware that allows distributed applications to securely share information in real time using a publisher/subscriber model) as the communication layer. This work addresses the need for robust methodologies.to simulate and evaluate the performance of complex electrical systems under distributed and real-time communication conditions.

The proposed architecture is organized into two main nodes: Node 1, which integrates the master subsystem (MS) where the generation and control system (CS) runs and where communication between Node 1 takes place, and Node 2, which handles the load (an electrical model (e.g., R-L-C) representing the consumption of a device; it is excited by the Thévenin equivalent source from Node 1 (V_MS) and returns the demand current (I_demand) to the co-simulation loop). Communication is established through a signal contract defined in two signal flows: flow 1 (Node 1→DDS) and flow 2 (Node 2→Node 1), transmitted as structured data types and managed with DDS (Data Distribution Service) policies configured to prioritize the most recent sample and avoid queues. For instrumentation, UDP↔DDS bridges were developed in Python, along with data logging and control scripts, ensuring traceability and capture of electrical and communication metrics documented in this report.

In summary, this work seeks to demonstrate the feasibility and performance of a distributed real-time co-simulation architecture based on RTI Connext DDS and OPAL-RT,

establishing experimental guidelines that allow quantifying the impacts of effective network latency and sampling granularity on both temporal and electrical metrics, and offering a replicable basis for future studies of electrical system integration in distributed environments.

Keywords: Distributed co-simulation, Real-time, OPAL-RT, RTI Connext DDS, Electrical systems

2 Introduction

Real-time distributed co-simulation has become established as an effective strategy for analyzing electrical systems [1] complex when software and hardware components run on heterogeneous and geographically separated platforms. In this context, OPAL-RT platforms for real-time simulation and MATLAB/Simulink environments allow for high-fidelity representation of electrical subsystems [2] and control, while middleware (intermediate software that facilitates communication and data exchange between distributed communication applications) such as RTI Connext DDS provide low-latency data exchange with explicit QoS control (a set of policies and parameters that allow you to configure how communication behaves in a middleware) [3][4].

This thesis addresses the design, implementation, and evaluation of a distributed co-simulation architecture that integrates two distinct nodes, communicating via a Python software bridge that translates UDP (User Datagram Protocol) datagrams into DDS (Data Distribution Service) samples and vice versa. Messaging is organized into two streams defined by a signal contract that specifies variables, units, direction, and frequency. DDS QoS policies are configured to prioritize the most recent sample and avoid queues (Reliability = BEST_EFFORT; History = KEEP_LAST).

The architecture also includes time capture at node 1, i.e., a high-resolution timestamp recorded with a monotonic clock. in it. With this mark, the temporal metrics (effective latency, jitter and data loss) are estimated, which are related to the electrical metrics (Vrms, Irms, Pmean and FP). [5][6]

Methodologically, three distributed co-simulation scenarios are compared that allow isolating the cost of software messaging, incorporating hardware effects, and having a reference without distributed communication (details and parameters are detailed in Methodology).

2.1 Context

The increasing complexity of modern electrical systems demands methodologies that allow the study of their behavior under dynamic conditions and in distributed simulation environments. In particular, the integration of renewable sources, storage systems, advanced controllers, and nonlinear loads poses challenges in terms of stability, power quality, and timing coordination between components [7]. In contrast, real-time co-simulation emerges as a highly valuable tool, enabling interaction between models run on different platforms and the analysis of electrical phenomena [4][8]under realistic communication conditions.

In this context, OPAL-RT platforms have become benchmarks for real-time simulation of electrical systems, offering the possibility of emulating complete subsystems with integration steps on the order of microseconds [9][2] and hardware-in-the-loop interaction capabilities [9][10]. In parallel, MATLAB/Simulink continues to be the standard in modeling and simulation of electrical and control systems, making it an ideal environment to complement OPAL-RT in distributed configurations

The communication component between electrical equipment is equally relevant. The DDS (Data Distribution Service) standard, defined by the OMG [11], and in particular its RTI Connext DDS implementation, provides data-oriented middleware with quality of service policies[3]that allow operation under latency, loss, and synchronization constraints. This approach has been used in critical areas such as automotive, aerospace, and industrial, and its incorporation into the co-simulation of electrical systems ensures scalability [12] and explicit control of information exchange.

In this way, the combination of OPAL-RT, MATLAB/Simulink and RTI Connext DDS enables a distributed architecture that not only allows the analysis of electrical subsystems in real time, but also allows experimentation with different communication configurations and quantifying their impact on the temporal and electrical performance of the simulation [4][1].

2.2 Problem Statement

The analysis of electrical systems in centralized simulation environments presents limitations when it is necessary to represent distributed interactions or communication conditions close to real-time operation [13]. Given the increasing complexity of these systems, platforms that run the simulation on a single computer do not allow for the accurate quantification of the effects of latency, jitter, losses, and temporal misalignment that emerge when subsystems exchange data across networks [14][1] Consequently, distributed co-simulation architectures that explicitly model communication between electrical equipment are explored and adopted

In practice, these limitations hinder the validation of control strategies, the integration of hardware-in-the-loop, and the comparison between different execution frameworks (e.g., Simulink versus OPAL-RT). Furthermore, while middleware solutions are widely used in other critical areas (such as DDS in automotive and aerospace), their application in electrical co-simulation is not yet sufficiently documented or standardized [14].

The absence of clear experimental guidelines on how to configure sampling, QoS, and timing strategies limits the reproducibility of studies and reduces the ability to extrapolate results to industrial settings [1][15]. In the specific case of academic environments, there are also network restrictions such as multicast blocking in university infrastructures that force the use of alternative networks (e.g., cellular hotspots), which introduces additional communication conditions that must be quantified and documented [16][17].

Consequently, the need arises to implement and validate a real-time distributed co-simulation architecture that allows:

- Compare centralized and distributed execution scenarios.
- Quantify the impact of the communication period in temporal and electrical metrics.
- Establish replicable configuration and operating criteria for future research.

2.3 Motivation

The development of modern electrical systems increasingly demands tools that allow the integration of multiple domains—generation, control, communications, and loads—into a single simulation environment [3][12] Real-time distributed co-simulation emerges as an alternative that not only enables the validation of complex models under conditions close to real operation, but also offers a flexible framework for studying how communications influence the stability and quality of electrical results.

The choice of RTI Connext DDS as middleware is due to its widespread use in critical sectors where reliability and control of data exchange are fundamental, such as in automotive and aerospace systems. [2][10]Integrating this standard into the electrical field helps to close a knowledge gap and explore its applicability in environments where real-time requirements are equally demanding

Additionally, the incorporation of OPAL-RT allows Simulink's capabilities to be extended to real-time simulation, which is essential for hardware-in-the-loop testing and the validation of control algorithms under realistic conditions. [10]This integration represents an opportunity to develop research practices that are replicable and comparable with previous work in the literature.

In the academic field, this work is also motivated by the need to overcome the network limitations faced by university laboratories, where infrastructure restrictions force the design of alternative architectures and the rigorous documentation of their impact on co-simulation. [16][17] Thus, the research not only seeks to demonstrate the technical feasibility of the architecture, but also to offer criteria that support future implementations in similar environments

2.4 General Objective

The general objective of this work is to implement and characterize the performance of a real-time distributed co-simulation architecture based on RTI Connext DDS and OPAL-RT, comparing three scenarios: Simulink-Simulink (S1), where both ends run Simulink software; Simulink-OPAL (S2), where OP4200 hardware is incorporated as a real-time simulator; and a reference scenario (S_ref), in which all subsystems run on a single system to establish the baseline of expected behavior. Additionally, the impact of the communication period (Ts_comm, data transmission interval between nodes) on time and electrical metrics is evaluated

In particular, the temporal metrics analyzed are effective latency (estimated average arrival offset with a single clock (receiver) with respect to the nominal calendar), jitter (variation of said latency between successive samples) and packet losses (percentage of messages that do not reach the receiver); for their part, in the electrical metrics, Vrms and Irms correspond to the effective values of voltage and current, Pmean to the average active power as a temporal average of the instantaneous power and the power factor to the ratio between active and apparent power, which reflects the voltage-current alignment and, consequently, the efficiency of the energy exchange.

2.5 Specific objectives

To achieve the general objective, the following specific objectives are set:

1. Implement the UDP↔DDS communication bridges and establish the signal contract between the SM/SC and Load subsystems.
2. Run the simulations with four repetitions per case and measure temporal performance metrics: latency, jitter, and packet loss.
3. Calculate and analyze the electrical metrics (Vrms, Irms, Pmean and power factor) and relate them to the temporal results obtained.
4. Compare scenarios S1, S2, and S_ref, identifying advantages, limitations, and trade-offs associated with each configuration.

2.6 Scope and limitations

Scope:

1. The research focuses on the implementation and validation of a real-time distributed co-simulation architecture that integrates Simulink, OPAL-RT (model OP4200), and RTI Connext DDS
2. Three study scenarios are considered: S1 (Simulink-Simulink), S2 (Simulink-OPAL) and S_ref (load within SM).
3. The analysis includes:
    a. Temporal metrics: latency (average arrival offset measured with a single clock), jitter, and losses (percentage of samples not received).
    b. Electrical metrics: Vrms and Irms (RMS values), Pmean (average active power) and FP (power factor).
4. Time parameters: Ts_core = 0.5 ms (base step size of the SM simulation in OPAL 4200) and communication period between nodes Ts_comm $\in$ {0.5; 1; 1.5; 2.5 ms}. Each simulation lasts $\approx$40 s.

5. The results are intended to be replicable in academic settings and to serve as a methodological reference for future research.

Limitations:

1. Although the OP4200 and OP4512 equipment was available, only the former could be used in the simulations, due to a failure in the OP4512 unit. This situation restricts the validation to a single operating hardware unit
2. Communication took place over a restricted data network, using a 3G hotspot, due to multicast blocking in the university infrastructure. This factor affects the results and should be considered when extrapolating them to other environments.
3. The performance analysis considers only the QoS parameters configured in BEST_EFFORT + KEEP_LAST (a configuration that allows avoiding data queues), without exploring more advanced DDS policies.

3 Theoretical framework

3.1 Co-simulation in electrical systems

3.1.1 Definition and purpose

Co-simulation is understood as the coordinated integration of two or more simulators or execution platforms that exchange variables at well-defined coupling points, in order to study complex systems that span multiple domains [4] (electrical, control, and communications). Its purpose is to reproduce realistic interactions between subsystems, maintaining the fidelity of each tool within its domain and allowing analysis of how communications impact overall performance [18].

3.1.2 Relevant typologies

Offline and real-time configurations can be distinguished, as well as SIL (Software-in-the-Loop, where both the controller and the plant are simulated in software), HIL (Hardware-in-the-Loop, where real hardware is connected to a simulated model in real time) schemes, including CHIL (Controller-Hardware-in-the-Loop, in which a physical controller connected to the simulated plant is tested) and PHIL (Power-Hardware-in-the-Loop, in which real power hardware is coupled to a simulated model), when controllers or power plants are integrated into closed loops. Regarding coordination, there are master-slave and distributed orchestration approaches, and numerically, fixed or variable steps depending on stability and latency requirements [4].

3.1.3 Data exchange and synchronization

The exchange can be time-stepped (blocking or non-blocking) or event-driven. In distributed environments, the quality of the interaction is affected by latency, jitter, and packet loss. These factors impact signal coherence and, consequently, the resulting electrical metrics.

Instrumentation using timestamps in the registers allows for the estimation of latency and jitter [1][19].

3.1.4 Coupling Algorithms

Coupling algorithms seek to ensure numerical consistency at the boundary between subsystems. Among the known families are ITM (Ideal Transformer Method), DIM (Damping Impedance Method), and TLM (Transmission Line Modelling) [20], which differ in the trade-off variables and the handling of delays. The choice depends on the problem's stiffness, the latency budget, and the jitter tolerance[4][21].

3.1.5 Middleware for Real-Time Co-simulation

DDS (Data Distribution Service) is a data-oriented standard that offers automatic discovery and a set of quality of service (QoS) policies to control reliability, history, resources, and timing. In co-simulation scenarios, configurations such as BEST_EFFORT and KEEP_LAST allow prioritizing the delivery of the most recent sample[3]and avoid queue buildup when the network is the bottleneck. Discovery configuration may require adjustments on networks with multicast restrictions

3.1.6 Real-time electrical simulation tools

OPAL-RT provides deterministic execution of electrical models with integration steps on the order of microseconds and supports HIL via RT-LAB; its complementarity with MATLAB/Simulink facilitates partitioning between electrical and control subsystems, maintaining traceability and control of the simulation step [9][2].

3.1.7 From the general framework to the case study

Next, the concepts of the theoretical framework will be applied to the specific architecture developed.

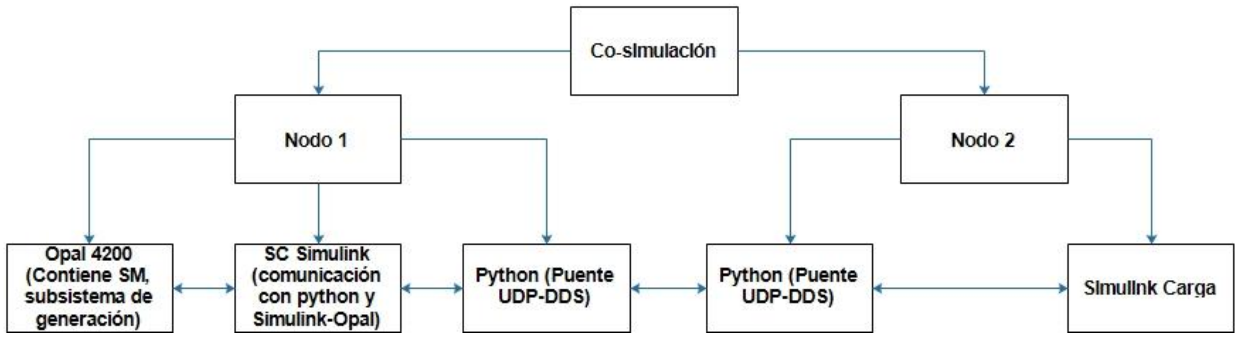

Figure3.1Co-simulation architecture

As shown in Figure 2.1, this thesis adopts a distributed architecture with Node 1 (SM and SC in Simulink) and Node 2 (Load) [16][17], these are coupled by a signal contract and communications bridge detailed in Chapter 4

3.2 DDS Middleware and RTI Connext

Data Distribution Service (DDS) is a middleware standard defined by the Object Management Group (OMG) that establishes a data-oriented publisher-subscriber communication model. Its main strength lies in the fact that it does not depend on a central server, but rather implements an automatic discovery mechanism among participants, which facilitates scalability in distributed systems and allows for the configuration of explicit quality of service (QoS) policies to control latency, reliability, and data persistence (OMG, 2015).

The commercial implementation of RTI Connext DDS has become established in critical sectors such as automotive, aerospace, and defense, due to its real-time optimized libraries and flexible QoS configurations (RTI, 2020). Among its most relevant policies for co-simulation environments are:

- Reliability: It can operate in RELIABLE or BEST_EFFORT mode. In electrical co-simulations with submillisecond integration steps, choosing BEST_EFFORT avoids queue congestion and prioritizes the delivery of the most recent sample.

- History: defines how many samples are retained in the reader buffer. Configuring KEEP_LAST with a low depth (depth=1 or 32 depending on the role) reduces accumulation and maintains data up-to-dateness.[3].
- Resource Limits: restricts the size ofmemoryand the number of samples in transit, adjusting to the capacity of the nodes and the network.

The use of DDS in distributed electrical systems has already been explored in experiences such as the Millapán project.[16]where coupling algorithms were applied alongside dynamic phasors. Furthermore, LAFAE's international experience demonstrates how middleware, when operating over wide area networks, introduces latency, jitter, and loss challenges that must be carefully evaluated in distributed simulations. These findings support the need to define QoS configurations adapted to the context of real-time co-simulation.

This report adopts RTI Connext DDS 7.5.0, configured with BEST_EFFORT and KEEP_LAST, to enable communication between Simulink, OPAL-RT, and the Python scripts. Discovery is restricted to Domain ID=0 and is complemented by UDP↔DDS bridges that filter and publish the signals defined in the communication contract (PingSM and PongN2). This integration allows for the precise quantification of how QoS decisions and sampling periods impact temporal and electrical performance metrics.

3.3 OPAL-RT and Real-Time Simulation

OPAL-RT platforms have been widely adopted in electrical systems research and development due to their ability to run models with integration steps on the order of microseconds, maintaining determinism and enabling hardware-in-the-loop (HIL) interaction. [9][2][10]. Through their RT-LAB environment, these platforms allow the deployment of models developed in MATLAB/Simulink and their execution on dedicated processors with FPGA acceleration [22], guaranteeing real-time execution under intensive loads [10].
Among its main advantages are:

- Deterministic execution: ensures that each modeling step is completed within the allotted time
- Partitioning flexibility: allows dividing a model into subsystems (SM, SC, load) distributed across different cores or equipment.
- HIL (Hardware-in-the-Loop) compatibility: integration of real hardware with a real-time model.
- Multiprotocol support: incorporates standard communication interfaces (Ethernet, CAN, FPGA I/O), facilitating interaction with other simulation platforms.

In the context of this research, OP4200 and OP4512 equipment were used. The former was used in the experimental campaigns, while the latter was available but not operational due to technical failures, which limited the tests to a single piece of hardware in service. This circumstance does not restrict the validity of the study, but it highlights the importance of documenting the condition of the equipment in laboratory environments.

The complementarity between OPAL-RT and MATLAB/Simulink is key to implementing distributed co-simulations: Simulink remains as a flexible modeling environment, while OPAL-RT executes the critical subsystems in real time. This integration allows for a more realistic evaluation of the performance of communication architectures, since they are tested in a physical environment with time and network constraints.

Previous works such as Millapán [16]have already demonstrated the feasibility of integrating OPAL-RT into co-simulation schemes with coupling algorithms, while international studies such as those by LAFAE have explored its use in geographically distributed simulation configurations, facing latency and stability challenges similar to those addressed by thismemory.

3.4 Coupling Algorithms (ITM, DIM, TLM)

In a co-simulation scheme, coupling algorithms determine how variables are exchanged at the boundary between electrical and control subsystems, and how the numerical stability of the overall system is ensured. Their design is fundamental to reducing integration errors when delays, losses, or variations exist in the communication steps [14]. Among the most relevant families are:

- **Ideal Transformer Method (ITM):** It is the most widely used reference algorithm due to its simplicity and robustness. It represents the coupling as an ideal transformer that ensures equality of coupling variables (current and voltage). It is recommended when delays are low and communication is relatively stable [4][21].
- **Damping Impedance Method (DIM):** It introduces a resistive damping term at the coupling boundary, which improves stability when high delays or jitter are present. Its main advantage is the reduction of numerical oscillations, although it can affect the fidelity of electrical variables in transient regimes [14][21].
- **Transmission Line Method (TLM):** models the coupling as a transmission line with distributed parameters, allowing the explicit representation of propagation delays. It is suitable for distributed simulations in geographically separated networks, although its implementation is more complex and requires greater computational power [4][21].

The selection of the algorithm depends on the communication conditions and the objectives of the study. In the case of this dissertation, the use of ITM is prioritized as the primary method [23], due to their stability in low-latency, high-sampling-rate scenarios. However, the value of DIM and TLM is recognized as alternatives for future research in scenarios with higher delays or in more unstable network infrastructures [24], such as those of a geographically distributed nature, studied in the laboratory at LAFAE [17].

3.5 Previous work and comparison with the literature

Real-time co-simulation of electrical systems has been the subject of various investigations, mainly focused on the validation of control schemes, the integration of renewable sources, and the evaluation of communication architectures. Palensky's studies stand out in the international literature [1], who establish the theoretical foundations of the co-simulation of intelligent power systems, including aspects of coupling, synchronization, and numerical stability

At the national level, the memory of Millapán [16] constitutes a direct precedent, by proposing a distributed co-simulation architecture that integrates OPAL-RT with coupling algorithms and dynamic phasors. Their work demonstrated the feasibility of applying ITM in an academic environment and provided relevant performance metrics, serving as a point of comparison for the present study

Internationally, in addition to the work of LAFAE, there is a large body of research on distributed co-simulation. Notable among these are orchestration frameworks such as HELICS, which is geared towards large-scale, multi-domain co-simulation [25], and FNCS, which integrates electrical simulators (e.g.,GridLAB-D) with communications simulators (ns-3) to study the impact of latency and losses[26], as well as compositing platforms like Mosaic for scenarios with changing topologies [27]. Interface improvements and stability/accuracy analyses have been published in real-time distributed co-simulation (GDS) [28], along with scaled applicability studies[29]. Network impacts on protection and control applications are reported in HIL/co-simulations with non-ideal networks [30] To mitigate the effects of network delay, several works use dynamic phasors and associated tools such as DPsim [31][32] Taken together, these works support the idea that latency, jitter, and losses affect the coupling of remote simulators and that strategies and frameworks exist to assess and mitigate them

In comparison to this background, the present report is distinguished by:

- Integrate RTI Connext DDS as communication middleware, specifically configured with BEST_EFFORT + KEEP_LAST policies, which are still poorly documented in the electrical context.
- Explicitly evaluate the relationship between temporal metrics (latency, jitter and losses) and electrical metrics (Vrms, Irms, Pmean, FP), establishing a quantitative bridge between network performance and electrical behavior.
- Documenting the impact of real network restrictions in academic environments (multicast blocking and 3G hotspot usage) adds a practical dimension to the analysis of distributed architecture.

In this way, the work is positioned as a contribution that complements Millapán's advances[16]and LAFAE [17], providing experimental evidence on the integration of DDS in real-time electrical co-simulation, and establishing replicable guidelines for future research at the University of O'Higgins and in international contexts

4 Working methodology

The methodology describes, sequentially, how the distributed co-simulation was designed, executed and analyzed to evaluate its temporal and electrical performance in S1, S2 and S_ref.

BASE PARAMETERS AND ASSUMPTIONS:

Definitions.

- Ts_core = "internal step" of the simulator (OPAL-RT/SM).
- Ts_comm = "how often" DDS is published/received between nodes
- Ts_core (DC) = 0.5 ms (SM step in OPAL-RT).
- Ts_core (AC) = 1 ms (observed stable configuration)
- Ts_comm (DC) $\in$ {0.5; 1; 1.5; 2.5} ms.
- Ts_comm (AC) = 1 ms (0.5 ms is discarded due to losses in preliminary tests).
- Duration per run $\approx$ 40 s.
- QoS DDS: Reliability = BEST_EFFORT; History = KEEP_LAST.
- Time traces: t_wall_ns (monotonic clock) recorded on node 1 Python for each sample sent/received, to estimate latency, jitter and losses.

PREPARATION AND INSTRUMENTATION:

Objective: to leave the test bench ready and observable.

Procedure:

- Load Simulink models (SM, SC, Load) and compile/run SM in OPAL-RT (OP4200).
- Configure RTI Connext DDS: types, topics (PingSM/PongN2) and QoS (BEST_EFFORT + KEEP_LAST).
- Enable the UDP↔DDS bridge (Python) and the publish/subscribe and log scripts (including the clock).
- Verify connectivity between nodes and DDS discovery.

SIMULATION EXECUTION

Objective: To generate comparable and repeatable data.

Procedure:

- Ordered startup: OPAL-RT / Simulink -> UDP bridge<->DDS -> publishers -> subscribers
- Run ~40s per run (scenario x factor).
- Register separate CSV files per observation point (e.g., Local_.csv, Opal_.csv, Simulink_.csv) with the columns: t(s), V_SM, I_SM, P_SM, I_dem.
- Maintain a constant network condition (e.g., hotspot or LAN) during each run.
- Record the value of Ts_comm used in each run.

SYNCHRONIZATION AND ALIGNMENT

Objective: to ensure temporal comparability between records.

Procedure:

- Alignment between files using cross-correlation (adjusts the timings to analyze everything when the exchange between nodes begins) (xcorr) using an electrical signal (I_SM when it is greater than 0, as it reflects the consumption of node 2).
- Verify effective duration.

METRICS CALCULATION

Temporary metrics per file (derived from the local registry):

- Losses (%):$Loss = \left(1 - \frac{N_{real}}{N_{esp}}\right) \times 100$
- Jitter $= J_{T_s} = \sqrt{\frac{1}{(N-1)-1} \sum_{i=2}^{N} (\Delta t_i - T_s^{comm})^2}$
- Effective latency $= L_{\text{eff}} = \frac{1}{N} \sum_{i=1}^{N} (t_i - [t_1 + (i-1)\, T_s^{comm}])$

Electrical metrics (same window, after cropping and alignment):

- Effective voltage $= V_{rms} = \sqrt{\frac{1}{N} \sum_{k=1}^{N} \left(V_{SM,k}\right)^2}$

- Effective current $= I_{rms} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(I_{SM,k}\right)^2}$
- Average active power $= P_{mean} = \frac{1}{N}\sum_{k=1}^{N} P_{SM,k}$
- Power factor $= FP = \frac{P_{mean}}{V_{rms} \cdot I_{rms}}$

GLOSSARY

- $\boldsymbol{T_s^{\mathrm{comm}}}$ = Nominal communication period (configured in the protocol).
- $\boldsymbol{N}$ = Number of samples received in the analyzed window
- $\boldsymbol{N}_{\mathrm{real}}$ = Actual number of samples received (in practice, N)
- $\boldsymbol{N}_{\mathrm{esp}}$ = Expected number of samples according to the nominal period $\boldsymbol{T_s^{\mathrm{comm}}}$
- $\boldsymbol{t_i}$ = Local arrival time of sample i at Node 1 in seconds.
- $\boldsymbol{t_1}$ = Local arrival time of the first window sample.
- $\boldsymbol{\Delta t_i = t_i - t_{i-1}}$ = Arrival interval between consecutive samples.

COMPARISON AND SENSITIVITY

Objective: to identify the effects of architecture and communication

Analysis:

- S1 vs S2: Impact of hardware and on sampling jitter, losses, and electrical RMSE
- S_ref vs S1/S2: deviation introduced by the distribution/network in Vrms, Irms, Pmean and FP.
- DC case: sensitivity of sampling jitter and losses versus Ts_comm and its correlation with RMSE and electrical metrics.
- AC case: effect of frequency (with Ts_comm = 1.0 ms) on the same metrics.

5 Solution Architecture and Development

The architecture proposed for this research is organized around two main co-simulation nodes, connected via RTI Connext DDS-based communication middleware, with intermediate bridges implemented in Python (UDP↔DDS). This arrangement allows for the evaluation of real-time interaction between electrical and control subsystems running in different environments.

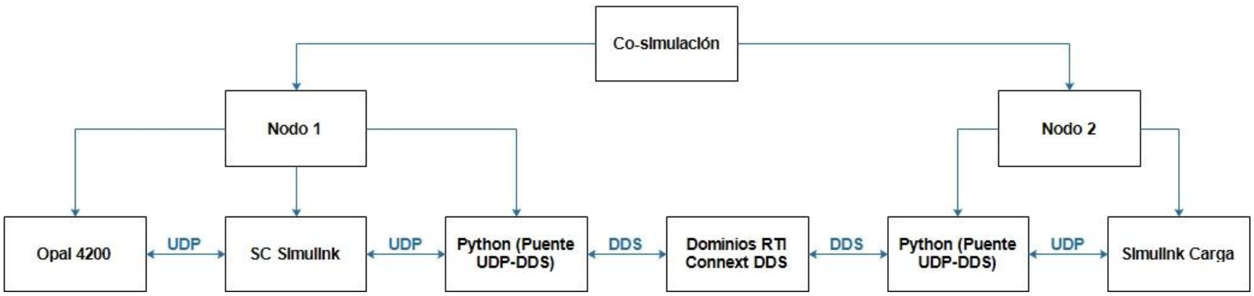

Figure5.1Co-simulation architecture

Explanation of figure 4.1:

- Node 1: integrates the master subsystem (SM) in the Opal 4200 and the control subsystem (SC-PC1-Simulink-OPAL-RT), both modeled in MATLAB/Simulink R2019a. This node constitutes the main source of the system's reference and control signals.
- Node 2: runs the load subsystem on PC 2, modeled in MATLAB/Simulink 2025. Its role is to receive signals from Node 1, calculate the load response and forward feedback information.
- OPAL-RT OP4200: was used to run the master subsystem (SM) in the S2 scenario, guaranteeing real-time operation with a fixed simulation step (Ts_core = 0.0005 s).

Communication between nodes is organized around a signal contract:

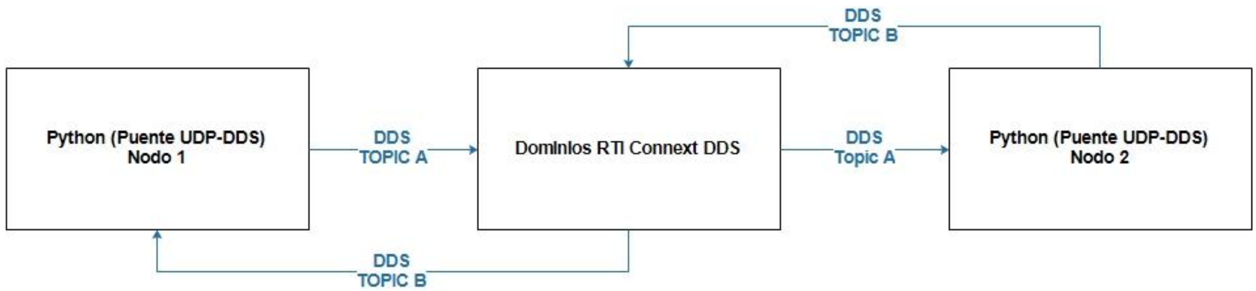

Figure5.2Signal diagram between nodes with RTI Connext DDS

As can be seen in Figure 4.2:

- TOPIC A = PingSM (Node 1→DDS): I_demand, V_SM, P_SM, I_SM.
- TOPIC B = PongN2 (Node 2→Node 1): I_demanda_eco2, I_demanda.

The data is published and subscribed to through topics A and B, configured under Domain ID = 0 (Global Domain). The adopted quality of service policy was BEST_EFFORT for reliability and KEEP_LAST for maintenance [3] For history, with typical depths of 1 (takes the most recent sample and avoids queues, allowing prioritizing the delivery of the most recent sample and reducing data accumulation; if not configured, any variation in the network will accumulate timing lags in data delivery and sending

At the network level, communication was carried out via a 3G hotspot due to multicast blocking in the university infrastructure. This introduced additional latency and jitter, which were documented and analyzed in the results.

The architecture (Figure 4.1) was tested in three experimental scenarios:

- S1 (Simulink-Simulink): Node 1 and Node 2 executed on different PCs, running all Subsystems in Simulink, i.e., the opal did not participate in this test as shown in figure 4.1 and SM was executed on PC 1 from Simulink.
- S2 (Simulink-OPAL): Complete architecture (Figure 4.1), i.e., SM integrated into the opal.
- S_ref (Reference): All subsystems within a single computer modeled in Simulink, serving as a reference to distributed scenarios.
- This organization allowed for a controlled comparison of the effect of distribution, the inclusion of OPAL-RT, and communication conditions on time and electrical metrics.

5.1.1 Detailed explanation of node 1

Node 1 houses the master subsystem (MS) and the control subsystem (CS). Both were implemented in MATLAB/Simulink R2019a to take advantage of its specialized libraries for electrical systems and digital control.

- Master subsystem (SM): represents the power source and generates the main system variables (V_SM, I_SM, P_SM). In scenario S2, this subsystem was executed directly on the OPAL-RT OP4200, ensuring a real-time integration step of Ts_core = 0.0005 s, allowing validation of the closed-loop hardware-software interaction.
- Control subsystem (CS): implements the monitoring and regulation routines for the sent and received variables. In all scenarios, it remained running in Simulink, ensuring flexibility to modify control strategies and capture data directly.

Node 1 acts as the primary publisher within the DDS architecture. Through topic A, it transmits the information grouped in the message (see Figure 4.2) PingSM to the middleware: I_demand, V_SM, P_SM, and I_SM. These values are sent to Node 2, where they are processed as payload inputs.

Likewise, Node 1 subscribes to topic B, receiving the feedback data grouped in the PongN2 message: I_demanda_eco2 and I_demanda. This information allows the communication loop to be closed and provides real-time feedback to the SM and SC variables.

Python scripts were implemented to implement the node, performing the functions of publishing, subscribing, and logging data in binary format. This intermediate layer ensures compatibility between Simulink, OPAL-RT, and DDS, and also allows for detailed control of the message flow.

5.1.2 Detailed explanation of Node 2

Node 2 (PC2) corresponds to the load subsystem, implemented in MATLAB/Simulink 2015. Its function is to receive the signals sent from Node 1, calculate the load response, and return feedback information to the master system.

The communication of this node was implemented through two complementary processes, supported by scripts in Python 3.13:

- N2SUB.py (DDS→UDP): subscribes to topic A, receiving the values published from Node 1 (PingSM: I_demanda, V_SM, P_SM, I_SM). This data is then forwarded via UDP to Simulink, through local port 25000, where it is processed as inputs to the load model.
- N2PUB.py (UDP→DDS): receives data from Simulink via UDP (port 21001), corresponding to the feedback signals (PongN2: I_demanda_eco2, I_demanda). It then publishes this information to topic B for Node 1 to process in real time.

In this way, Node 2 acts as both consumer and producer within the DDS architecture: it consumes the signals from the SM and SC to execute the load dynamics, and produces feedback signals that allow the distributed simulation loop to be closed.

5.2 Signal contract

The signal contract explicitly defines the variables exchanged between nodes in each simulation cycle. This agreement is essential to ensure compatibility between models in Simulink, OPAL-RT, and the publish/subscribe processes in DDS.

The definition was organized into two data flows (see figure 4.2):

5.2.1 PingSM (Node 1 → DDS)

Node 1, through the script N1PUB.py, publishes the following variables to the topic DatosSMTopic:

- I_demanda: demand current requested from the source.
- V_SM: voltage delivered by the master subsystem.
- P_SM: active power generated by the master subsystem.
- I_SM: current measured in the master subsystem.

These signals are consumed by Node 2 using N2SUB.py, which forwards them to Simulink as inputs for the load model.

5.2.2 PongN2 (Node 2 → Node 1)

Node 2, using the N2PUB.py script, publishes the feedback variables to the DatosN2Topic topic:

- I_demanda_eco2: demand current reflected in Node 2.
- I_demanda: demand current calculated by the load model.
- This data is received by Node 1 through N1SUB.py, closing the communication loop and allowing the SC to adjust the system variables in real time.

The signal contract was implemented using the CoSimTypes.idl file, which was processed by the RTI code generator (rtiddsgen) to produce the Python data structures (CoSimTypes_PingSM and CoSimTypes_PongN2). This ensures that transmitted messages adhere to a common format, regardless of the source or destination platform.

5.3 DDS communication configuration

Communication between the co-simulation nodes was implemented using the RTI Connext DDS 7.5.0 middleware, with a configuration adapted to the low-latency and high-frequency transmission requirements of the analyzed scenarios. The configuration was based on three main elements: topics and data types, QoS and queuing policy, and Domain ID and discovery.

5.3.1 Topics and data types

Two main topics were defined:

- DatosSMTopic (Topic A), associated with the PingSM flow (Node 1 → DDS), which includes the variables I_demanda, V_SM, P_SM and I_SM.
- DatosN2Topic (Topic B), associated with the PongN2 flow (Node 2 → DDS → Node 1), which carries the variables I_demanda_eco2 and I_demanda.

Both topics were generated from the CoSimTypes.idl file, which allowed for the compilation of compatible data structures in Python and maintained consistency between publishers and subscribers.

5.3.2 QoS and queuing policy

A quality of service (QoS) policy was adopted aimed at minimizing latency and avoiding the accumulation of obsolete samples:

- Reliability: configured in BEST_EFFORT mode, prioritizing the delivery of the most recent sample over the strict reliability of each package.

- History: set to KEEP_LAST, which ensures that only the latest samples are stored according to the configured depth.

This scheme was chosen to prevent communication from being penalized by network delays, while maintaining consistency with the real-time nature of the simulation.

5.3.3 Domain ID and discovery

All DDS participants were configured with Domain ID = 0, ensuring they were part of the same logical communication session. The peer-to-peer discovery mechanism was used in standard mode. However, the university network infrastructure presented multicast traffic limitations [33], which forced the use of a 3G hotspot as an alternative connection method. This aspect, although limiting, allowed for the evaluation of the impact of a network with real latency and jitter conditions on the co-simulation

6 Results

Following the experimental objectives defined in Chapter 3, the results of the real-time distributed co-simulation architecture are presented below. Two sets of tests were conducted: first, the temporal and electrical performance was evaluated by varying the communication period (DC scenario), and subsequently, the system's response to alternating current signals was analyzed by varying the frequency (AC scenario). In all cases, the three defined configurations were compared: S1 (Simulink-Simulink co-simulation), S2 (Simulink-OPAL-RT co-simulation), and S_ref (local reference scenario), as outlined in the specific objectives. Each experiment was run for approximately 40 seconds. The findings are presented below in a clear and sequential manner, emphasizing narrative coherence and connection to the study objectives

6.1 Results with variation of the communication period (Tests in DC)

First, the architecture's performance was analyzed under direct current (DC) conditions, varying the communication period (Ts_comm). These tests allow us to characterize the influence of data exchange granularity on temporal performance metrics (latency, jitter, and packet loss) and on the electrical quality of the simulation (voltage, current, power, and power factor), as proposed in the experimental objectives.

Simulation with Ts = 0.5 ms

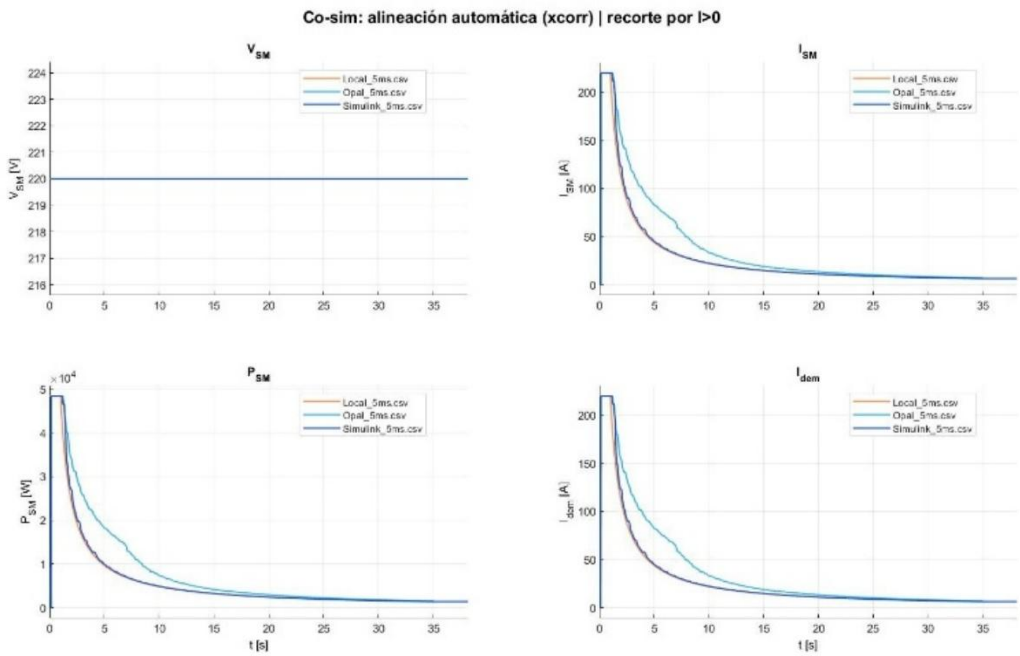

Figure6.1Global image of the signals in the simulation with Ts_comm = 0.0005 sec

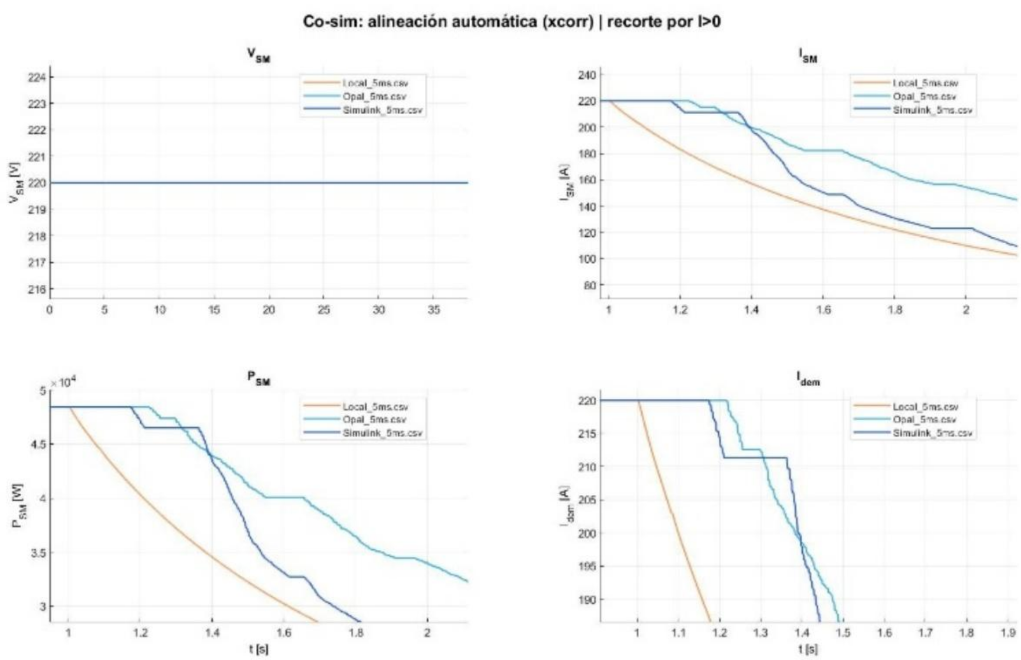

Figure6.2Enlarged image of the signals in the simulation with Ts_comm = 0.0005 sec

Figures 5.1 and 5.2 show the typical waveforms of the co-simulation for a communication period Ts_comm = 0.5 ms. Figure 5.1 presents the overall system response (voltage and current) in the distributed scenario versus the local reference scenario, while Figure 5.2 shows an enlarged detail of the same signals. It can be observed that the output of the co-simulation model closely follows the local reference. Only a slight phase shift or delay is noticeable in the signal from the remote node, attributable to the interface bridge (SC) implemented in the OPAL-RT 4200, which does not support asynchronous blocks and adds a small conversion delay. This aspect will be examined in more detail later, when discussing AC testing

Scenario with Ts = 2.5 ms

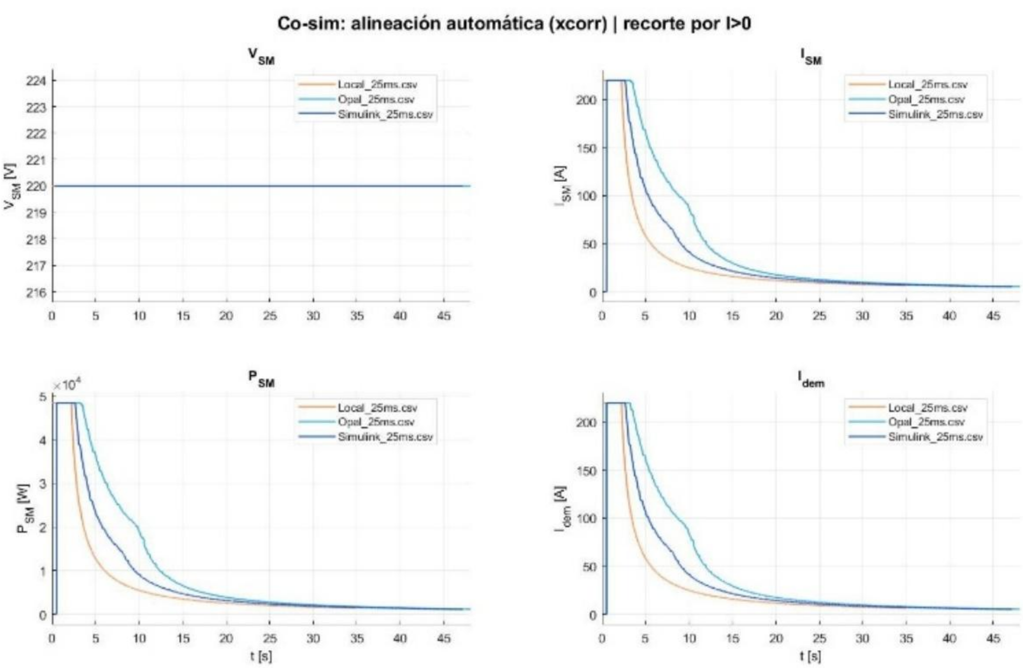

Figure6.3Global image of the signals in the simulation with Ts_comm = 0.0025 sec

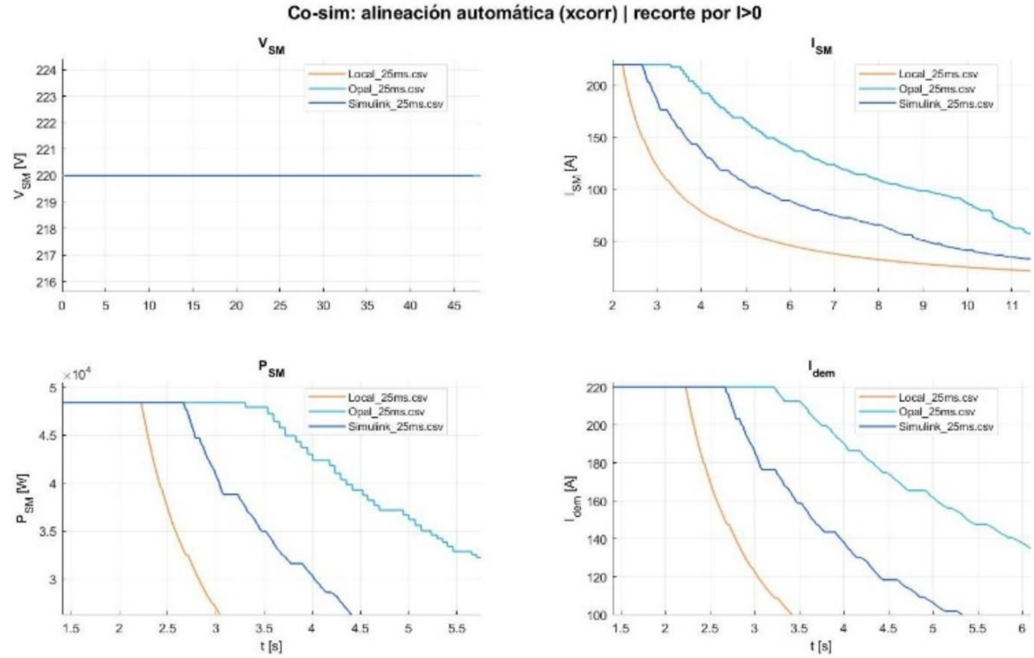

Figure6.4Enlarged image of the signals in the simulation with Ts_comm = 0.0025 sec

Figures 5.3 and 5.4 illustrate the case with Ts_comm = 2.5 ms, again showing the global signals and their amplification. In this more relaxed communication scenario, the co-simulation also manages to replicate the behavior of the reference system, although a slightly larger accumulated lag than in the previous case can be observed, consistent with the longer data exchange period.

Table6.1Temporal Metrics in DC

Ts_comm	Scenario	Effective Latency (ms)	Jitter (ms)	Losses (%)
0.5 ms	S_ref (Local)	0.631	0.526	0.00
	S2 (OPAL)	0.363	0.466	**36.73**
	S1 (Simulink)	0.363	0.447	**35.45**
1.0 ms	S_ref (Local)	1.049	0.483	0.00
	S2 (OPAL)	1.039	0.470	0.00
	S1 (Simulink)	1.032	0.465	0.00
1.5 ms	S_ref (Local)	1.531	0.294	0.00
	S2 (OPAL)	1.525	0.290	0.00
	S1 (Simulink)	1.514	0.268	0.00
2.5 ms	S_ref (Local)	2.547	0.345	0.00
	S2 (OPAL)	2.545	0.331	0.00
	S1 (Simulink)	2.550	0.319	0.00

Table 5.1. Time metrics measured in scenarios S_ref, S1, and S2 for different values of Ts_comm (0.5–2.5 ms). Effective latency, jitter (standard deviation of latency), and packet loss percentage are listed for each case:

- Ts_comm = 0.5 ms: Effective latency ≈0.36 ms in both S1 and S2 (close to the configured value of 0.5 ms), with high jitter around 0.45–0.47 ms (on the order of the communication period itself). Very high packet losses were recorded: approximately 35.5% in S1 and 36.7% in S2, while S_ref (since it does not involve network communication) showed no noticeable losses or latency.

- Ts_comm = 1.0 ms: Effective latency ≈1.04 ms on S1 and S2, coinciding with the configured period of 1 ms. Jitter was reduced to ~0.47 ms, less than 50% of Ts_comm. No packet loss (0%) occurred in any of the distributed scenarios, demonstrating stable communication at this rate.

- Ts_comm = 1.5 ms: Effective latency ≈1.53 ms, consistent with the period. Jitter decreased to ~0.29 ms (≈19% of Ts_comm). No losses (0%) were observed in S1 or S2.
- Ts_comm = 2.5 ms: Effective latency ≈2.54–2.55 ms in both distributed scenarios. Jitter remained at ~0.32–0.35 ms, significantly lower than the period (≈13% of Ts_comm). No packet loss (0%) was recorded with this communication period.

Analysis of temporal metrics: Based on the results above (Table 5.1), several important observations can be drawn about the temporal performance of the co-simulation:

- **Effective latency:**In all co-simulation cases, the measured latency was very close to the configured communication period, confirming that the architecture effectively operates in real time (meeting the overall objective of maintaining real-time exchanges). For example, with Ts_comm = 1 ms, an effective latency of ~1.04 ms was obtained, and with Ts_comm = 2.5 ms, around 2.54 ms. This indicates that the DDS middleware and communication bridges successfully transmit data with a delay consistent with the desired frequency. It is worth noting that at Ts_comm = 0.5 ms, although the effective latency (~0.36 ms) was lower than the period (because many packets were not transmitted before the next cycle), the channel could not sustain that transmission rate, as evidenced by the high losses. In summary, latency remained aligned with Ts_comm in all scenarios, validating the real-time operation of the platform, although in the extreme case of 0.5 ms this configuration is inapplicable due to the instability it generates.

- **Jitter:**Jitter (variability of effective latency between consecutive packets) showed a decreasing trend as Ts_comm increased. With the shortest period (0.5 ms), jitter was on the order of 0.45–0.52 ms, a value practically equal to the transmission interval and which contributed significantly to the observed losses (the high variability prevented all messages from being processed on time). As the period increased, jitter decreased both relatively and absolutely: for example, with Ts_comm = 1 ms it was reduced to ~0.47 ms

(≈47% of the period), with 1.5 ms it dropped to ~0.29 ms (≈19% of the period), and with 2.5 ms it remained at ~0.32 ms (barely ~13% of the period). This indicates that longer communication periods provide more time slack to absorb variations in transmission, stabilizing the link In Ts_comm configurations ≥ 1 ms, the jitter was low enough not to compromise synchronization between nodes, while at 0.5 ms its magnitude, comparable to the period itself, severely compromised communication stability (contributing to packet loss). In conclusion, jitter is acceptable starting at 1 ms, while at 0.5 ms it exceeds what is tolerable for reliable operation.

- **Package loss:**The packet loss rate proved to be the critical metric that defined the architecture's operating regime. With Ts_comm = 0.5 ms, massive losses were observed in the distributed scenarios: approximately 35% in S1 and 37% in S2. This high loss explains the significant deviations in the electrical metrics for that case (as will be seen later) and demonstrates that the system cannot sustain such a high communication rate given the network limitations and the QoS configuration used. In contrast, for Ts_comm = 1.0, 1.5, and 2.5 ms, no losses were recorded (0% in all cases): communication was completely reliable in both S1 and S2, confirming that above a certain threshold (~1 ms), the architecture is capable of maintaining data exchange without saturating the channel. In conclusion, there is a stability threshold around Ts_comm = 1 ms for this platform: below that period, the required transmission rate causes network saturation and considerable losses; above (or equal to) 1 ms, the transmission is sustainable and loss-free, thus fulfilling one of the specific objectives of identifying the minimum stable operating conditions of the co-simulation.

Analysis of electrical metrics Having evaluated the time-domain metrics, we proceed to examine the main electrical metrics obtained in these DC tests: RMS voltage (Vrms), RMS current (Irms), average active power (Pmean), and power factor (PF). These variables allow us to quantify the fidelity with which the co-simulation reproduces the electrical behavior of the system compared to the local reference, and to detect how communication conditions (e.g., losses or delay) can affect the system's response, as outlined in one of the experimental objectives. The results for S1 and S2 in relation to S_ref are described below:

- **Voltage (Vrms):**The RMS voltage remained constant at 220 V in all scenarios and for all communication periods tested. In both the base case (S_ref) and S1 and S2, Vrms = 220 V without appreciable variations, even in the case with high losses (0.5 ms). This indicates that the voltage magnitude was not affected by the geographical distribution or communication irregularities, expected behavior given that the voltage bus was controlled (master subsystem) and communication primarily affects the current demand. In summary, Vrms behaved as a robust variable under the co-simulation conditions, remaining unchanged and consistent with the reference model in all cases

- **Current (Irms):**The RMS current value showed greater sensitivity than voltage to communication imperfections and the inclusion of real-time hardware. In the reference scenario S_ref, the RMS current ranged between approximately 53 A and 66 A depending on the system's operating point in each test, with these values serving as an ideal reference. In S1 (Simulink-pure Simulink), Irms remained close to the reference when communication was stable, but experienced slight reductions under loss conditions. Specifically, with Ts_comm = 0.5 ms (where there was approximately 35% loss), Irms = 50.8 A was measured in S1, approximately 7% below the reference value (≈54.5 A). This decrease is consistent with the fact that packet loss reduces the current demand transmitted to the load node. At Ts_comm = 1 ms (0% losses), Irms in S1 rose to 52.6 A, practically coinciding with the reference (~53 A); and for 1.5 ms and 2.5 ms, Irms was 64.8 A and 64.1 A respectively, values also aligned with S_ref (which was around 65 A in those cases). In contrast, scenario S2 (Simulink-OPAL) showed an opposite trend, with an overestimation of the current compared to the reference. With Ts_comm = 0.5 ms, S2 yielded Irms = 62.3 A, approximately +15% higher than the local value. This unexpected increase suggests that data loss and processing in OPAL introduced artifacts that increased the measured current. At Ts_comm = 1 ms, Irms in S2 was 58.8 A (about +5-6 A above ~53 A in S_ref); At 1.5 ms, the current rose to ~73.0 A (vs. ~65 A ref); at 2.5 ms, it reached 77.9 A (vs. ~64 A ref). That is, in S2, the effective current was consistently higher than in the reference, and this gap widened as Ts_comm increased. In summary, S1 accurately reproduces the system current as long as the communication does not introduce losses, while S2 tends to overestimate the current delivered to the load. This effect is attributable to the internal dynamics of the OPAL-RT and its way of processing information in real time (for example, possible accumulations or delays in the UDP↔DDS interface). This difference between distributed scenarios fulfills the objective of identifying the discrepancies introduced by the hardware in the quality of the electrical simulation.

- **Active power (Pmean):**The average active power supplied to the load clearly reflected the impact of the communication conditions. At S_ref, Pmean remained at values of 6.3-8.0 kW depending on the operating point (which varied slightly with Ts_comm due to the master generator's control dynamics). In scenario S1, Pmean results were very close to the reference in all cases without losses, while in the presence of losses, an underestimation was evident. At Ts_comm = 0.5 ms, Pmean = 5.93 kW in S1, slightly lower than the S_ref value (~6.30 kW), consistent with the current reduction due to lost packets. As Ts_comm increased and losses were eliminated, the power recovered to the expected level: for example, with 1 ms, Pmean = 6.29 kW vs. 6.30 kW in the reference (virtually identical). With 1.5 ms, the power was approximately 8.06 kW vs. 8.02 kW; and with 2.5 ms, it was approximately 7.65 kW vs. 7.92 kW, differences of less than 5%. This confirms that S1 delivers the same active power as the local system as long as the communication does not introduce discontinuities. S2, on the other hand, showed higher power values than the reference in all cases, similar to what was observed in the current. For Ts_comm = 0.5 ms, Pmean = 8.33 kW in S2, which represents approximately +24% over the approximately

6.7 kW of S_ref in that case. With 1 ms, the power in S2 was approximately 7.25 kW vs. 6.30 kW in S_ref (+15%); with 1.5 ms, it increased to 9.46 kW vs. 8.02 kW (+18%). With a 2.5 ms response time, it reached 9.92 kW vs. 7.92 kW (+25%). Clearly, S2 systematically overestimates the active power delivered, and this overestimation is amplified with longer Ts_comm times. In other words, integrating the OPAL-RT into the co-simulation introduces a positive bias in the power measurement, likely linked to the higher simulated current and possible delays in the control feedback. Thus, while S1 maintains power according to the reference except when data is missing (losses), S2 exhibits different electrical performance that should be considered when evaluating the architecture's effectiveness.

- **Power factor (PF):**The power factor, defined as the ratio of active to apparent power, provides an integrated indicator of how the system delivers energy in both magnitude and phase. In the reference scenario S_ref, the power factor remained between 0.53 and 0.56 in the various tests, reflecting the load and control conditions of the base system. For S1, the power factor behaved virtually the same as in S_ref as long as communication did not introduce disturbances: with Ts_comm = 1 ms, power factor = 0.544 vs. 0.535 in S_ref; with 1.5 ms it was 0.566 vs. ~0.56; with 2.5 ms, 0.543 vs. ~0.54. The only deviation occurred at 0.5 ms, where power factor = 0.531 in S1, slightly lower than the (approximate) 0.58 of the local case, due to the reduction in Pmean caused by packet losses. These results confirm that scenario S1 maintains a power factor virtually identical to the reference under normal conditions, meeting the expectation of electrical fidelity. In contrast, S2 showed higher power factors than the reference in all configurations, consistent with the previously noted overestimation of current and power. For example, at Ts_comm = 0.5 ms, S2 obtained a power factor of 0.608 (vs. ~0.56 in S_ref); at 1 ms it was 0.560 (vs. 0.535); at 1.5 ms, 0.589; and at 2.5 ms, 0.579. That is, S2 exhibits a consistently higher power factor than the actual one, indicating that it delivers proportionally more current per unit of active power than it should. This deviation is attributable to the real-time processing of OPAL-RT and its interaction with Ts_comm, which slightly alters the phase or magnitude of the simulated current. In conclusion, S1 preserves the power factor of the original system, while S2 tends to deviate towards higher PFs, which is a significant effect to consider when integrating loop hardware.

Comparison between scenarios and the effect of Ts_comm: By globally comparing the evaluated scenarios (S1, S2 vs. S_ref), the advantages and limitations of distributed co-simulation under different communication configurations are identified, as one of the specific objectives sought. In terms of time metrics, both distributed schemes showed qualitatively similar behavior: with Ts_comm = 0.5 ms, neither achieved sustainable transmission (losses >35% in both cases), while from Ts_comm = 1 ms onwards, communication was stable and lossless. The difference lies in the fact that S2 introduced an additional fixed delay associated with the OPAL-RT (visible in the signal figures). In terms of electrical metrics, the differences were more pronounced: S1 achieved values very close to the reference whenever Ts_comm ≥ 1 ms (lossless), demonstrating that the software-to-software distribution can replicate the original behavior with high fidelity. S2, on the other hand, exhibited a systematic deviation: higher current, higher power, and therefore higher power factor than expected. This divergence was accentuated with longer Ts_comm times due to the accumulation of delays and the nature of the real-time hardware. This consistent overestimation suggests that the presence of the OPAL-RT imposes a different dynamic (possibly filtering or internal delays) that slightly alters the system's response compared to the purely Simulink model.

Comparing each scenario with the reference (S_ref), it is confirmed that S_ref, lacking network communication, provides the ideal baseline against which the others are compared. At Ts_comm = 0.5 ms, both S1 and S2 deviated significantly from S_ref (S1 underestimating and S2 overestimating) due to massive losses and delay, confirming that this period is not viable for proper coupling. At Ts_comm = 1 ms (the first lossless case), S1 replicated virtually all of S_ref's metrics (effective latency and matching electrical metrics), while S2 showed notable differences in current and power. For Ts_comm values of 1.5 and 2.5 ms, both scenarios maintained transmission stability (0% losses), but S2 deviated increasingly from S_ref in electrical terms as the period increased, while S1 maintained good coherence with the reference. This reinforces the conclusion that S1 is the most suitable scheme for reproducing the original behavior, and that S2, while introducing hardware-in-the-loop realism, entails compromises in the accuracy of the results.

From the perspective of the impact of Ts_comm on architecture, the key findings are summarized as follows:

- With Ts_comm = 0.5 ms, the data rate exceeds the system capacity → the channel becomes saturated with high losses and large misalignments (making co-simulation unfeasible in this configuration).
- With Ts_comm = 1.0 ms, a point of equilibrium is reached: communication is reliable (0% losses), the effective latency aligns with real time, and in S1 the electrical metrics are virtually identical to the reference. S2 also operates lossless at 1 ms, but already shows slight overestimations of Irms/Pmean, indicating the first differences introduced by OPAL.

- With Ts_comm = 1.5–2.5 ms, transmission remains stable and lossless. In the case of S2, electrical deviations are amplified (current and power are increasingly overestimated). S1 remains close to the local case, demonstrating robustness even with higher communication delays.

In summary, the DC test results show that the distributed co-simulation architecture is fully functional starting at Ts_comm ≈ 1 ms, meeting the expectations for stable real-time operation. Below this threshold, the communication configuration becomes unsustainable due to network saturation, while above 1 ms the system operates reliably, albeit with a larger delay. Scenario S1 (both subsystems in distributed Simulink) proved capable of reproducing the behavior of the reference system with high fidelity, provided communication is adequate, thus achieving the objective of validating the software-to-software architecture. On the other hand, the S2 scenario (OPAL-RT integration) provided real-time hardware realism but introduced systematic deviations in the electrical metrics (fulfilling the objective of identifying limitations and compromises of hardware inclusion): S2 showed higher values for current, power, and power factor compared to the local case, especially accentuated with long Ts_comm, which should be taken into account in applications where the accuracy of these magnitudes is critical.

6.2 Results with frequency variation (Tests in AC)

Additionally, alternating current (AC) tests were performed to evaluate the impact of signal frequency on the stability and fidelity of the distributed co-simulation. In this set of experiments, the communication period was set to a value proven to be stable (Ts_comm = 1 ms, according to the result of the previous section), and the frequencies of the electrical signals generated by the master system were varied. Specifically, a frequency sweep was performed on the sinusoidal source of the master generator with values of {1, 2, 3, 5, 7, 10, 20, 30, 40, 50} Hz, keeping all other parameters constant (including the integration step of the models, Ts_core = 0.0005 s). Each frequency was tested both in the local scenario (S_ref, with both subsystems on the same node) and in the distributed co-simulation scenario. Since preliminary tests showed that Ts_comm = 0.5 ms was unfeasible due to losses (this value was discarded for AC), Ts_comm = 1 ms was used in all AC tests, thus ensuring a reliable channel to focus the analysis on the effects of frequency. In accordance with the experimental objectives, these tests allow for the quantification of temporal and electrical metrics as a function of the operating frequency, thereby evaluating the performance limits of the distributed architecture in a full-wave dynamic scenario.

During AC testing, the S2 (Simulink-OPAL) configuration was found to have severe limitations even at low frequencies: at 1 Hz, the additional delay introduced by the OPAL-RT resulted in a very large phase shift and a substantially attenuated current response (as detailed later), compromising the validity of that scenario. Therefore, the main analysis focuses on the S1 (Simulink-Simulink) co-simulation, which proved to be more stable. However, occasional references to the performance of S2 at 1 Hz will be included to highlight these limitations.

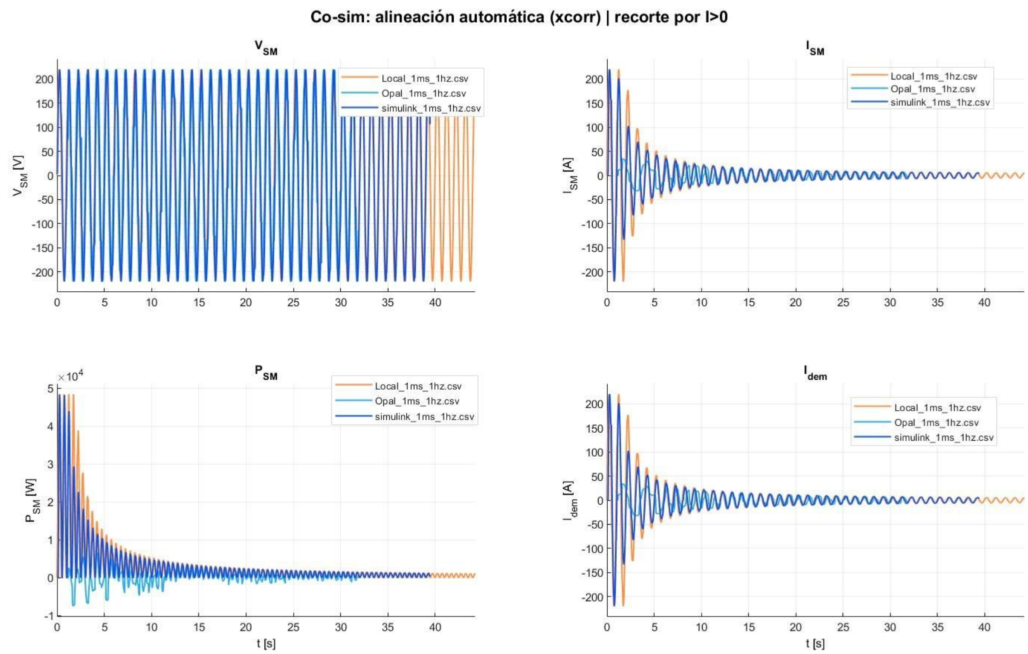

Figure6.5Global AC signals for f = 1 Hz

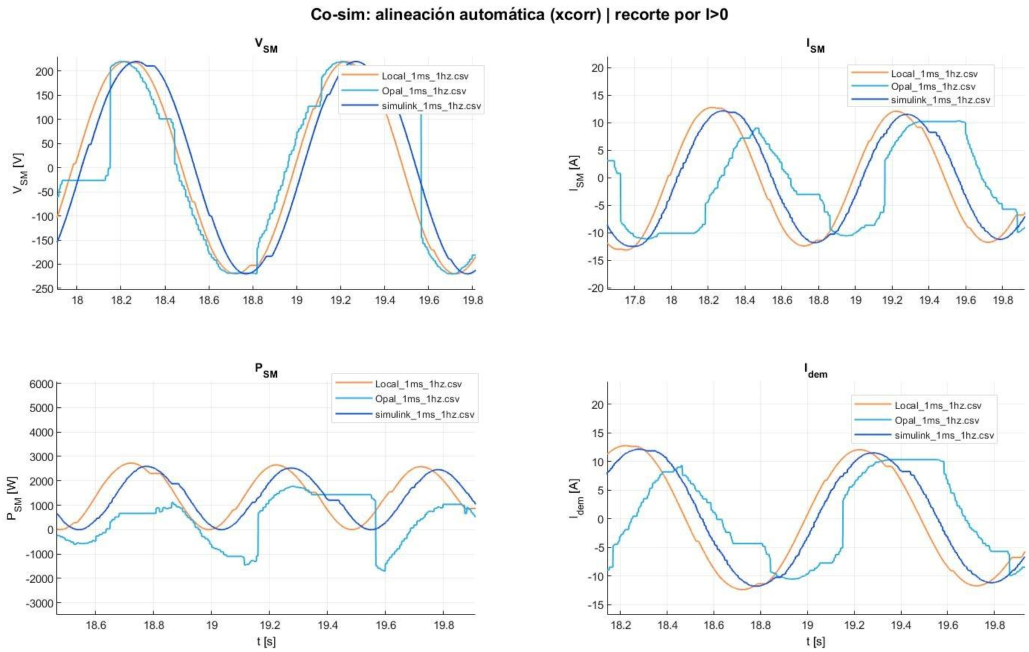

Figure6.6Enlarged detail of the AC signals for f = 1 Hz.

Figures 5.5 and 5.6 illustrate the voltage and current signals obtained for a frequency of f = 1 Hz in both the local reference scenario and the co-simulation (S1). Figure 5.5 shows the complete waveforms over several cycles, and Figure 5.6 presents a detailed zoom of one cycle, allowing the alignment between the two to be observed. It can be seen that at 1 Hz the co-simulation reproduces the response of the local system almost exactly: the voltage and current curves practically overlap, with only a small, constant phase shift caused by the communication.

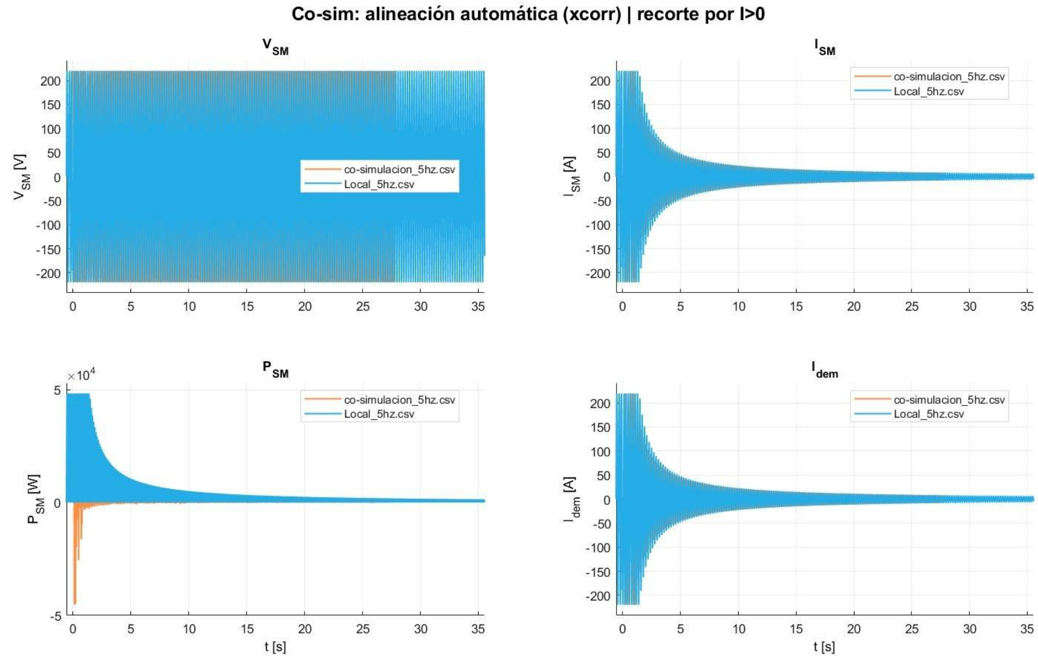

Figure6.7Global AC signals for f = 5 Hz.

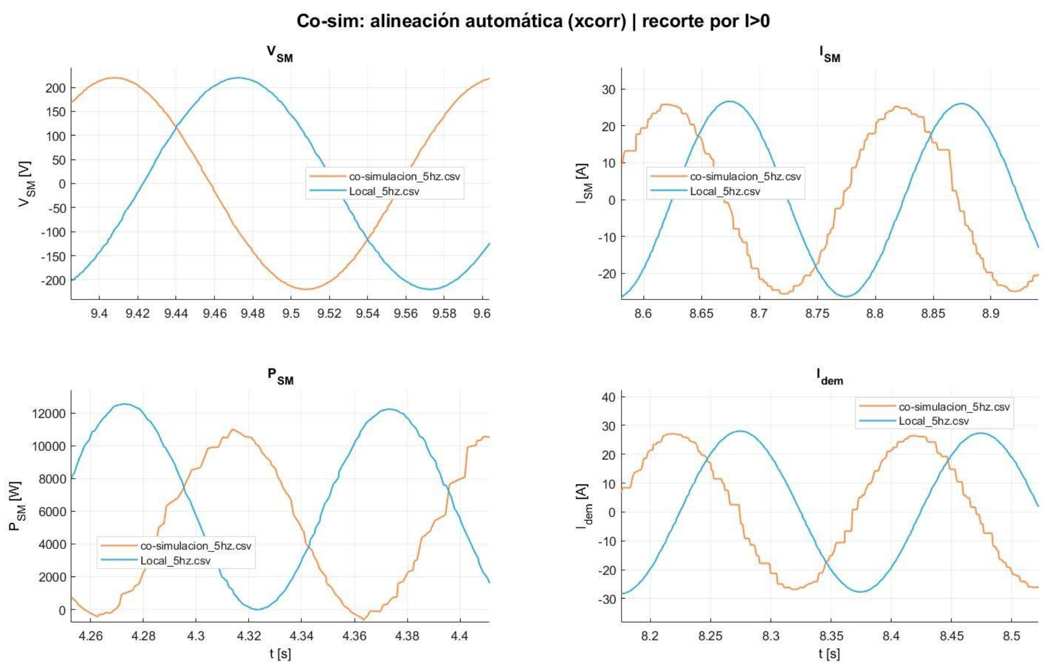

Figure6.8Enlarged detail of the AC signals for f = 5 Hz

Figures 5.7 and 5.8 present the case f = 5 Hz (global and detail, respectively), where a slight difference between the current signals of the distributed vs. local scenario is already beginning to be perceived: the amplitude and phase of the current in S1 deviate slightly from S_ref, indicating the first signs of degradation due to the fixed delay of 1 ms which, although small in absolute terms, represents a larger percentage of the period of the 5 Hz signal.

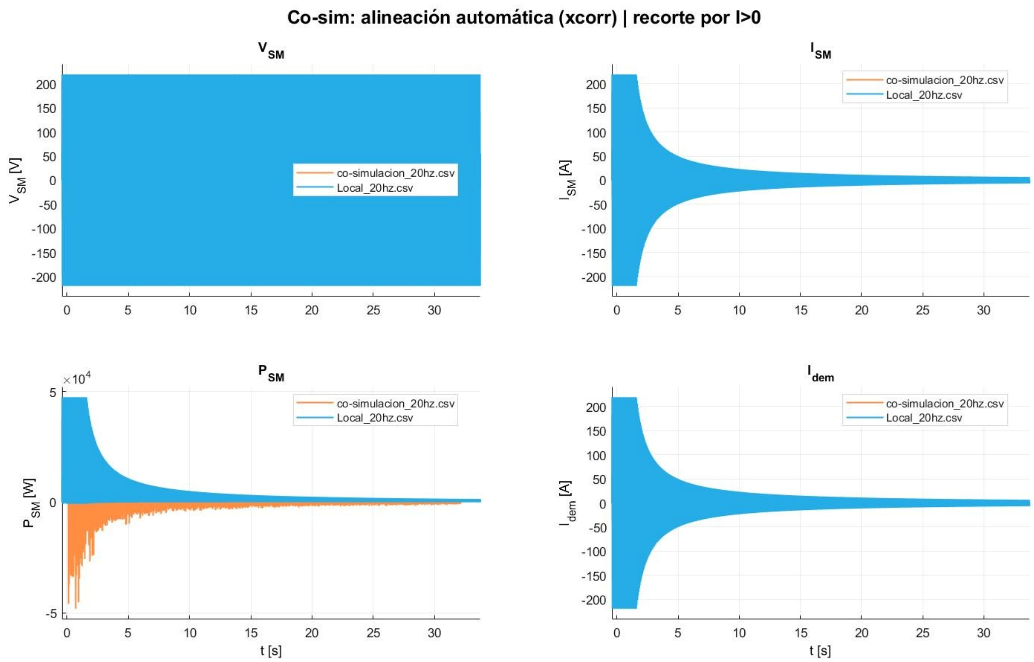

Figure6.9Global AC signals for f = 20 Hz

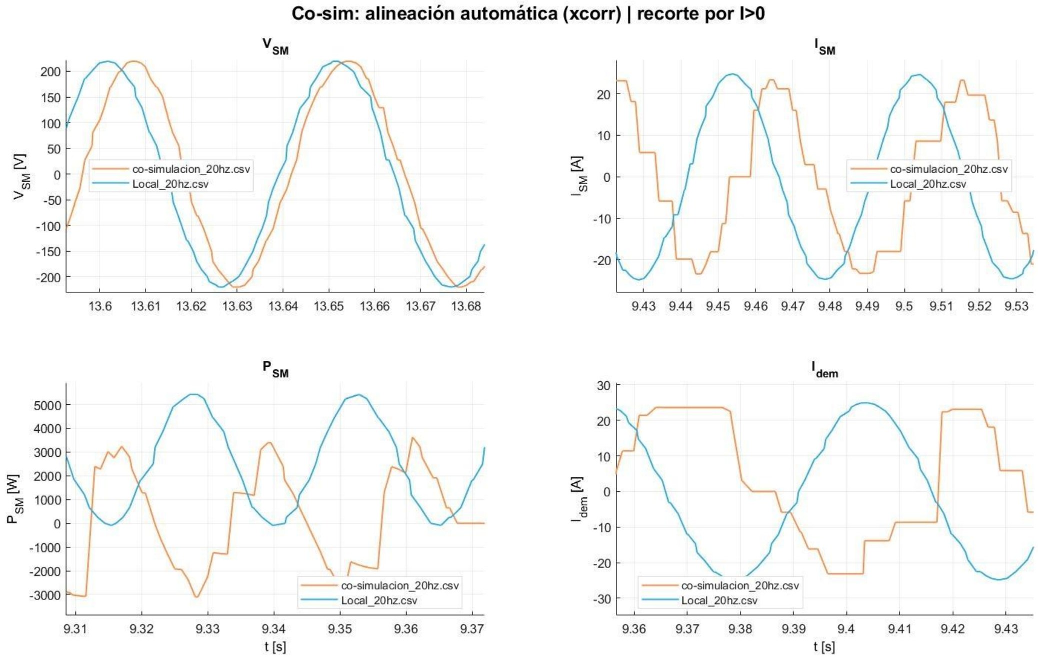

Figure6.10 Enlarged detail of the AC signals for f = 20 Hz

Finally, Figures 5.9 and 5.10 show the results for f = 20 Hz. In this case, a considerable phase shift between the response of S1 and the reference is evident: the current in the distributed scenario is noticeably out of phase with the voltage, and even its amplitude and waveform are altered, anticipating the drastic degradation that will occur at even higher frequencies (close to 50 Hz). These figures visually illustrate the trend: as the operating frequency increases, the fidelity of the co-simulation deteriorates, thus fulfilling the expectation that the fixed communication latency imposes a practical limit on the frequency of signals that can be simulated correctly

The following section details the time metrics measured in the AC tests. At all frequencies considered, the effective latency remained around 1.04 ms, consistent with the configured Ts_comm = 1 ms (i.e., communication continued to operate in real time). Jitter ranged from 0.45 to 0.55 ms depending on the frequency; in absolute terms, this jitter is small, but in relative terms (as a fraction of the AC signal period) it becomes more significant at higher frequencies.

For example, 0.5 ms of jitter represents only 1% of the period of a 1 Hz signal (1000 ms), but is equivalent to 2.5% of the period of a 20 Hz signal (40 ms) and 5% of the period of a 50 Hz signal (20 ms). This implies that latency variations that are acceptable at a fixed magnitude can compromise the relative synchronization of fast signals.

The accumulated time offset, meanwhile, increased with frequency in scenario S1: although Ts_comm is constant, higher frequencies generate more samples per second, allowing for the observation of larger net offsets during the simulation interval. Finally, regarding packet loss, almost zero results were obtained at most frequencies with Ts_comm = 1 ms: the communication channel proved to be reliable under these conditions. Only marginal losses were detected in three specific cases: 1 Hz (S1, with ≈1.85% packet loss), 2 Hz (S1, ≈0.84% loss), and surprisingly at 50 Hz in the Local case (~1.78% loss). This last value in S_ref could be attributed to some limitation in data recording or to the fact that, at 50 Hz, the local system experiences small irregularities in signal sampling (these are not communication losses, but perhaps omitted sampling points when calculating metrics). In any case, losses in S1 were essentially 0% for 3-50 Hz, confirming that the architecture maintained the reliability of the communication channel with Ts_comm = 1 ms in the analyzed frequency range. In summary, from a temporal perspective, the AC tests indicate that the communication link continues to operate adequately (real-time latency and no significant losses) even as the system dynamics frequency increases; however, jitter becomes increasingly critical at high frequencies, which foreshadows problems with the fidelity of the electrical variables. In fact, it can be preliminarily concluded that, although the communication infrastructure does not collapse at 50 Hz, the quality of the co-simulation is indeed affected by the fixed delay compared to the signal period.

Table6.2AC Temporal Metrics

f (Hz)	Scenario	Effective Latency (ms)	Jitter (ms)	Losses (%)
1	Local (S_ref)	1.036	0.447	0.00
	S2 (Opal)	1.047	0.482	0.00
	S1 (Simulink)	1.036	0.502	1.85
2	Local (S_ref)	1.073	0.552	0.00
	Co-sim	1.048	0.541	0.84
3	Local (S_ref)	1.043	0.502	0.00
	Co-sim	1.045	0.496	0.00
4	Local (S_ref)	1.114	0.585	0.00
	Co-sim	1.037	0.493	0.00
5	Local (S_ref)	1.079	0.559	0.00
	Co-sim	1.047	0.531	0.00
7	Local (S_ref)	1.082	0.558	0.00
	Co-sim	1.034	0.476	0.23
10	Local (S_ref)	1.050	0.497	0.00
	Co-sim	1.039	0.451	0.00
20	Local (S_ref)	1.132	0.583	0.00
	Co-sim	1.050	0.457	0.00
30	Local (S_ref)	1.095	0.562	0.00
	Co-sim	1.048	0.487	0.00
40	Local (S_ref)	1.094	0.574	0.00
	Co-sim	1.033	0.437	0.00
50	Local (S_ref)	1.033	0.491	1.78
	Co-sim	1.047	0.516	0.00

This conclusion is confirmed by examining the electrical metrics obtained in the frequency sweep. Table 5.2 summarizes the time metrics for each frequency and scenario; the corresponding electrical quantities were calculated from this table and the signal recordings. In all AC tests, Ts_comm = 1 ms was used, so the comparison between frequencies only reflects the effect of system dynamics in the presence of a fixed delay of ~1 ms. The key findings are described below, emphasizing the fidelity of the co-simulation (S1) with respect to the local scenario and the observed limitations. For clarity, the observations are grouped by frequency range:

Low frequencies (1-3 Hz): In this range, the co-simulation faithfully reproduces the behavior of the local system. The differences in electrical metrics between S1 and S_ref were less than 5% in all cases. For example, at 1 Hz, virtually the same RMS current and active power were obtained in S1 as in the reference (within the normal dispersion between repetitions), and at 3 Hz, maximum coherence between scenarios was recorded, with no signs of degradation. This indicates that for slow dynamics (long periods >> Ts_comm), the 1 ms communication delay is negligible, and the distributed architecture operates with high fidelity.

Mid-range frequencies (5-10 Hz): Significant deviations begin to appear in the co-simulation scenario starting at approximately 5 Hz. In the 5 Hz test, for example, the average active power at S1 dropped by approximately 20% compared to the local case, and the power factor decreased from approximately 0.54 at S_ref to approximately 0.47 at S1. This suggests that the small 1 ms delay begins to introduce a noticeable phase shift between voltage and current, reducing the power effectively transferred to the load. At 10 Hz, this trend intensifies: a power factor of approximately 0.41 was measured at S1 compared to approximately 0.54 at S_ref, indicating an even greater loss of synchronization (the current becomes more out of phase with the voltage, decreasing the in-phase component responsible for the active power). Although the RMS voltage and current values at S1 were still on the order of those of the local case, the power delivery was considerably underestimated compared to the reference. In summary, in the 5-10

Hz range, the fidelity of the co-simulation begins to degrade, with increasing errors in current, power, and power factor as the frequency increases.

High frequencies (20–50 Hz): In this range close to the grid frequency (50 Hz), the degradation of the co-simulation response was severe. At 20 Hz, the S1 co-simulation showed a drastic reduction in delivered active power: the average Pmean value in S1 was barely a fraction of that obtained in S_ref, and the power factor dropped almost to zero (≈0.017), indicating that the current was almost in quadrature with the voltage (phase shift close to 90°) due to the delay. For frequencies of 30 Hz and above, the situation was completely reversed: negative active power and an inverted power factor appeared in S1. This means that, from the perspective of the master generator, the distributed load returned power at certain times, a physically artificial result caused by the phase shift of more than 90° between the simulated voltage and current. At 30 Hz, power factors (PF) of approximately −0.3 were observed, and at 40–50 Hz, PF ~ −0.4 in S1. In contrast, the local scenario consistently maintained positive power levels and PF ~ 0.5–0.54 across the entire range. These results demonstrate that the distributed architecture, with a fixed delay of 1 ms, cannot adequately handle high frequencies: the delay introduces a significant phase shift every half cycle, ultimately inverting the voltage-current phase relationship. In other words, at frequencies on the order of the fundamental frequency of electrical systems (~50 Hz), the distributed co-simulation ceases to be operational, confirming the prediction of the full-wave architecture's inoperability at these frequencies.

In AC operation, even small effective delays can manifest as phase shifts between voltage and current as frequency increases. This effect explains why the power factor decreases markedly (and even becomes negative) at high frequencies, not attributing the result to an intrinsic electrical phenomenon of the model, but rather to the observable time shift in the arrival of samples.

Consequently, a partial conclusion can be drawn for the AC tests: the distributed architecture maintains electrical fidelity only up to approximately 5 Hz. Below or around this frequency, the latency/jitter effect (1 ms) is small enough not to significantly distort the electrical magnitudes; however, at higher frequencies, the communication-induced phase shift produces increasing errors in power delivery, even reversing the apparent power flow, which is unacceptable. In practical terms, this means that the proposed real-time co-simulation is appropriate only for relatively slow phenomena (e.g., second-long transients or sub-synchronous fluctuations), while for phenomena in the 10-50 Hz range (fundamental waves or power system harmonics), improvements in synchronization or drastic reductions in latency would be required to obtain reliable results. This finding fulfills the objective of evaluating the operational limits of distributed co-simulation, explicitly linking temporal performance with the quality of the electrical simulation.

Next, scenarios S1, S2, and S_ref are briefly compared in the context of AC tests to complete the analysis in accordance with the experimental objectives:

Local (S_ref): As expected, the local reference scenario maintained consistent performance across the entire frequency range. Vrms remained at approximately 155-156 V in all tests (a slight difference from the 220 V of the DC tests, possibly because the RMS value depends on the AC operating point). Irms, Pmean, and PF at S_ref remained stable and consistent at each frequency, reflecting only the load's inherent variation at different frequencies, without grid effects. In particular, the power factor at S_ref remained close to 0.54-0.56 for all frequencies, indicating that the base model's behavior did not change except with the excitation frequency. This scenario served as an ideal baseline for identifying deviations introduced by the distribution.

Simulink-Simulink Co-simulation (S1): At low frequencies ($\leq$3 Hz), S1 reproduced the local scenario with high fidelity, fully achieving the purpose of co-simulation. From 5 Hz onward, S1 began to differ from the local case: drops in Irms and Pmean relative to S_ref were observed, along with a sustained decrease in the power factor as frequency increased, as previously described. These differences grew with frequency, becoming very noticeable at 10 Hz and critical at 20 Hz and above. However, it is important to note that up to ~5 Hz, S1 co-simulation performs adequately, which can be useful for studies of slow dynamics or control in distributed systems. Beyond this point, however, the results of S1 deviate significantly from reality (S_ref), indicating the need for caution or compensation if higher-frequency phenomena are to be simulated using this architecture.

Simulink-OPAL Co-simulation (S2): This scenario exhibited difficulties from the very beginning of the frequency sweep. Even at 1 Hz, S2 showed an excessive response delay: the recorded time was ~1.1 s (compared to ~0 s in S_ref and ~0.03 s in S1), which implies that the OPAL signal was out of phase by more than a full cycle with respect to the reference. Consequently, the electrical metrics in S2 at 1 Hz were significantly attenuated; for example, Irms ≈ 12 A in S2 versus ~40 A in S_ref, demonstrating that a large portion of the current demand was not being transferred correctly in the co-simulation with OPAL. This extreme underestimation of current and power led to S2 being discarded as a valid scenario for AC analysis. In fact, S2 simulations at higher frequencies were not performed, since the performance at 1 Hz already indicated its infeasibility (the OPAL-RT could not even keep up with 1 cycle per second under the given configuration). In summary, the inclusion of the OPAL-RT in the AC tests was unsuccessful, revealing a significant limitation of this device or its integration into the architecture: for AC signals, the delay introduced by the hardware can be too large compared to the signal period, invalidating the co-simulation. This result fulfills the objective of identifying the limitations of the S2 scenario and suggests that optimizations (e.g., clock synchronization between OPAL and Simulink, use of asynchronous blocks, or reducing the computational load) would be necessary for the OPAL-RT to participate in co-simulations of faster phenomena.

In summary, the AC tests allowed us to determine the effective dynamic range of the distributed co-simulation architecture. It was found that, with a 1 ms delay, network communication imposes no restrictions up to certain limits (no losses up to 50 Hz, constant effective latency), but it does introduce a fixed phase shift that is negligible at very low frequencies and critical at high frequencies. In practical terms, at higher frequencies, a fixed delay manifests as a voltage-current phase shift that degrades the power factor (PF), hence the very low or even negative values observed between 20 and 50 Hz. The architecture is therefore suitable for low-frequency phenomena (on the order of 1-5 Hz) but rapidly loses fidelity beyond this threshold, unless the latency is further reduced or the phase shift is compensated. Thus, the experimental objective of evaluating the influence of frequency on co-simulation was met: it is now known that, under current conditions, ~5 Hz constitutes the practical limit for acceptable distributed electrical simulation.$\boldsymbol{T}_s^{\mathrm{comm}} =$

7 Discussion

The observed temporal metrics, such as effective latency, jitter, and packet loss, allowed for the evaluation of communication quality in the distributed co-simulation. In the tests performed, it was observed that a higher base effective latency in the communication link introduces additional delays in the interaction between subsystems, which can slow down the overall system response. Likewise, high jitter (high variability in delay) tends to momentarily desynchronize the arrival of data, generating irregular update intervals that complicate accurate tracking between the two parts of the simulation. Finally, the presence of packet loss, although low in most cases, indicates interruptions in the data flow that can result in jumps or inconsistencies in the exchanged signals. Overall, these results highlight that stable and fast communication (low latency, minimal jitter, no loss) is essential to maintain the temporal synchronization and fidelity of the real-time co-simulation

7.1 Relationship between Time Metrics and Electrical Results

Electrical results, including RMS voltages and currents (Vrms, Irms), average power (Pmean), and power factor (PF), were directly influenced by the quality of the communication timing. Slow or unstable communication affected the coherence between the voltage and current waveforms in the distributed subsystems. For example, when the effective latency increased significantly or jitter introduced unpredictable variations, effective phase shifts were observed between the voltage and current signals, reducing the power factor and potentially causing unwanted power flow (reactive power or even momentary reverse power). Similarly, in scenarios with high delays, certain measurements showed slight drops in Vrms and Irms compared to the ideal case, suggesting that the subsystems were unable to sustain the same voltage or current levels due to a delayed response. Despite these differences, when communication operated with low latency and controlled jitter, the electrical metrics in the distributed systems remained very close to those of the reference simulation, maintaining Vrms, Irms, and Pmean values within normal ranges and a virtually unity power factor. This confirms that the electrical integrity of the system can be preserved in distributed co-simulation as long as the communication conditions ensure adequate time synchronization.

7.2 Influence of the Communication Period and the Data Network

The simulations revealed that the communication configuration, particularly the communication step size (Ts_comm), and the characteristics of the data network used critically influence the co-simulation results. Regarding Ts_comm, a significant trade-off was found: an excessively small Ts_comm (for example, on the order of 0.5 ms) forces packets to be sent very frequently, which overloads the channel and results in significant packet loss during testing. This manifested as instability: the co-simulation could not be sustained under such short communication intervals due to network saturation and the processing limits of the nodes. In contrast, by using a slightly larger Ts_comm (in the range of 1-2.5 ms), data transmission stabilized and losses disappeared, demonstrating reliable communication. However, this increase in Ts_comm resulted in the subsystems operating more decoupled between each data exchange. In practice, this means that even without losses, each subsystem advanced a longer interval on its own before correcting its state with data from the other. This resulted in a gradual time misalignment between the outputs of both subsystems, which in S2 contributed to increasing divergences in long-term electrical measurements. These findings indicate that the optimal communication interval should be as small as possible without saturating the network: this minimizes delay and accumulated time divergence, while also avoiding losses.

Regarding the communication network, the results were conclusive. In all distributed runs (S1 and S2), a 3G hotspot wireless link was used exclusively; this medium introduced greater variability and delay compared to the local reference scenario (S_ref, without a network). Measurements show that under 3G, latency jitter was high and fluctuating, with episodes of sudden delay that degraded synchronization between nodes and caused sporadic losses. This behavior resulted in disturbances to the electrical signals: for example, small load oscillations that were handled smoothly in S_ref, under 3G in S1/S2 generated delayed and damped responses, affecting the stability of the exchanged power and the voltage-current alignment (power factor). Although a dedicated LAN/Ethernet was not tested, it is reasonable to project that a dedicated wired link would provide more consistent and lower latencies, further reducing jitter and virtually eliminating losses. With a higher quality network, both subsystems could operate with smaller Ts_comm without degrading time coherence even under high dynamic demands.

8 Conclusions and Future Work

This work confirmed the feasibility of a real-time distributed co-simulation of an electrical system, integrating software and hardware subsystems that operated on different computers but managed to reproduce, in general terms, the expected behavior of the complete system. The evidence gathered in Chapter 5 showed that the communication component is crucial for electrical fidelity: latency and, especially, its variability (jitter) determine the temporal coherence between the exchanged V_SM and I_SM signals. When this temporal coupling degrades, effective phase shifts emerge that alter Vrms, Irms, Pmean, and reduce the power factor (PF). Conversely, when communication operates with low latency, contained jitter, and no losses, the electrical metrics remain close to the local reference (S_ref), preserving the quality of the result.

Throughout the simulations, the communication step size (Ts_comm) emerged as a critical lever with a structural trade-off: values that are too small increase traffic and can saturate the network or nodes, while larger values alleviate the load but amplify the temporal decoupling between exchanges, favoring gradual divergences when a common clock is not present. The quality of the underlying network reinforces this diagnosis: wireless links (3G hotspot and local Wi-Fi) introduced latency spikes and jitter that affected synchronization between nodes, while comparative evidence suggests that a dedicated wired Ethernet LAN would allow for a sustained reduction in latency and its variability, enabling shorter Ts_comm steps without loss and improving temporal cohesion even in the most demanding scenarios.

Finally, the absence of a shared time reference explained part of the relative drift between nodes observed in more demanding scenarios. This result guides the conceptual conclusion: technical feasibility is demonstrated, but its quality remains contingent on

- A network infrastructure with deterministic behavior.
- Explicit time synchronization mechanisms, without which the benefits of optimizing Ts_comm are partially limited.

8.1 Future Work

In light of the findings, the immediate path forward is to harden the communications layer. In terms of infrastructure, it is proposed to migrate to a dedicated wired Ethernet network (isolated segment, switches with QoS, and, where supported, larger frames) to stabilize end-to-end latency and contain jitter. On this basis, it will be possible to reduce Ts_comm to the threshold where no losses appear, increasing coupling between subsystems without sacrificing stability. In parallel, it is pertinent to test stricter DDS QoS configurations (with depths adjusted to the operating regime), along with queue telemetry and sample deposit/ejection to diagnose localized saturations

The second key aspect is time synchronization. This involves incorporating IEEE 1588 PTP (in Boundary/Transparent Clock mode, where intermediate network clocks correct transit delays and accurately redistribute the time reference) [34][35] or a clock discipline with GNSS/GPS (which adjusts local clocks to the global satellite signal, ensuring a common absolute reference) [36] would allow sharing a common time reference, reducing relative drift and improving event traceability. Operationally, it is advisable to maintain the current clock in messages and logs, adding Tx/Rx stamps to estimate latency and jitter online, and implementing adaptive buffers that dynamically adjust waiting and resampling in response to network variations

A third area of focus concerns interface and control algorithms. Since the ITM method was sensitive to delay/jitter in certain regimes, it is proposed to compare its performance with more robust approaches such as DIM or TLM and to explore predictive strategies (e.g., feed-forward control or state estimators) that compensate for the effects of variable delay without compromising stability. These techniques can limit the impact of residual fluctuations even when the network is largely deterministic.

In AC operation, a natural extension is the incorporation of dynamic phasors as a representation of electrical states. By operating in a phasor space (e.g., dq/αβ or synchronized phasors), information exchange becomes less sensitive to small phase shifts and micro-variations in delay, facilitating the maintenance of V-I coherence with moderate Ts_comm and reducing bandwidth overhead compared to full-wave switching. This model change should be accompanied by comparative validation of metrics (Vrms, Irms, Pmean, FP) against the reference and S1/S2 cases, while maintaining consistent Ts_core and analysis windows.

Finally, it is suggested that the architecture be scaled to multiple nodes (generation, storage, and distributed loads) and subjected to representative events (stepped load changes, voltage dips, frequency variations), increasing the number of repetitions per case to report mean $\pm$ SD ($n=4$) and perform sensitivity analyses with respect to window and alignment. With these improvements, the platform will advance from demonstrated feasibility to robust operation in near-real-time industrial scenarios, where communication speed and precise synchronization are critical.

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10 Attachments

All files will be available at this link:https://drive.google.com/drive/folders/1cZ9r_E6i-eWDn15wwcgfrX4-XyUEFJce?usp=drive_link

10.1 Python Scripts

This appendix includes all the Python scripts used for co-simulation. They are listed with their main role, inputs and outputs, signals involved, and DDS configuration. Table 9.1-1 summarizes each script

Script	Primary Role	Inputs/Outputs (UDP)	Signals and Types	Topic/QoS DDS	Notes
CoSimTypes.py	Defines the DDS data structures (PingSM and PongN2)	—	I_demanda, V_SM, P_SM, I_SM / I_demanda_eco2, I_demanda (float)	—	Generated with rtiddsgen. Signal contract basis
Launch_N1.py	Launcher for Node1; opens N1PUB.py, N1SUB.py, and N1_csv.py in three consoles	—	—	—	Ask for file name and publication period, create CSV in logs/.
N1_csv.py	Block UDP→CSV Logger	UDP in 26001; creates CSV	I_demand, V_SM, P_SM, I_SM (double)	—	Write blocks to logs/<name>.csv.
N1PUB.py	DDS publisher with draining and pacing	UDP in 21000; UDP log 26001	Publish PingSM with the four signals (float32)	TopicSMTopicData; QoS: BEST_EFFORT + KEEP_LAST depth 1	Take the last valid packet and publish each Ts_comm.
N1SUB.py	DDS→UDP Subscriber for Node1	DDS in DataN2Topic; UDP out 25001	I_demand_eco2, I_demand (double)	Topic DataN2Topic; BEST_EFFORT + KEEP_LAST depth 1	Drain the DDS tail and forward only the last sample.
N2SUB.py	DDS→UDP Subscriber for Node2	DDS in DatosSMTopic; UDP out 25000	I_demand, V_SM, P_SM, I_SM (double)	TopicSMTopicData; BEST_EFFORT + KEEP_LAST depth 1	Send the latest packet to Simulink Node2.

N2PUB.py	UDP→DDS Publisher for Node2	UDP in 21001	Publishes PongN2 with I_demanda_eco2 and I_demanda (float32)	Topic DataN2Topic; BEST_EFFORT + KEEP_LAST depth 1	Drain the UDP buffer and publish the last valid value.
launch_N2.py	Node2 Launcher; opens N2PUB.py and N2SUB.py	—	—	—	Opens scripts in separate windows (Windows/Linux).

Each script has configurable environment variables (e.g., DDS_DOMAIN_ID, TS_COMM_MS, UDP_DEST_IP_N1, etc.), allowing ports and topics to be adapted to different test scenarios.

10.2 DDS Configuration

The simulations employ RTI Connext DDS. All topics use Domain ID = 0, Best Effort reliability, and a KEEP_LAST history policy with a depth of 1, preventing queues from accumulating. Data types are defined in CoSimTypes.idl and automatically generated (CoSimTypes.py). To reproduce the results:

- QoS: The default configuration was used except for the reliability and history parameters.
- rtps.ini file: specifies the Domain Participant, QoS profiles, and peer list; make sure to include it in the configuration folder.
- XML profiles: If custom profiles were used (for example, to vary depth or resource_limits), include them here.

10.3 Map of ports and topics

The following table summarizes the signal mapping between scripts, UDP ports, and DDS topics. It is included to clarify the information flow.

10.3.1 UDP ports and mapping to DDS topics

Component	UDP Inbound Port	UDP Outbound Port	Associated DDS Topic	Signals	Observations
N1 Publisher (N1PUB.py)	21000	26001 (optional logger)	DataSMTopic (PingSM)	Demand, VSM, PSM, ISM	Read 4 doubles; publish float32; send binary copy to the logger.
N1 Subscriber (N1SUB.py)	— (DDS in)	25001	DatosN2Topic (PongN2)	I_demanda_eco2, I_demanda	Forwards via UDP in 16 KB (2 doubles).
N1 Logger (N1_csv.py)	26001	—	—	Demand, VSM, PSM, ISM	Save CSV files with timestamps (t_wall_ns).
N2 Publisher (N2PUB.py)	21001	— (DDS out)	DatosN2Topic (PongN2)	I_demanda_eco2, I_demanda	Antiqueues: publishes only the last packet.
N2 Subscriber (N2SUB.py)	— (DDS in)	25000	DataSMTopic (PingSM)	Demand, VSM, PSM, ISM	Send 32 B (4 doubles) to Simulink Node 2.

10.4 Simulink Models (DC and AC)

This appendix contains screenshots and descriptions of the Simulink models used in scenarios S1 (Simulink-Simulink), S2 (Simulink-OPAL), and S_{ref} (embedded load in SM). Two .slx files are included:

- rampa05ms.slx: Basic DC model with load ramp and base sampling time of 0.5 ms. Shows the interaction of SM, SC and off-grid load.
- rampa_co_AC_1ms.slx: AC model for validation, with a 50 Hz reference signal. Includes active power measurement and phase calculation.

Each model is described by means of a technical sheet (solver, Ts_core, Ts_comm, relevant blocks and initial conditions).

10.5 OPAL-RT platform and software

To reproduce the co-simulation, the following were used:

- Hardware: OPAL-RT OP4200/OP4512; specify exact model and capabilities (number of cores, FPGA capacity, etc.).

- Software: MATLAB R2019a; RT-LAB 2020.4.1.166; Python 3.13; RTI Connext DDS 7.5.0.

- Configuration: Using the same versions ensures consistency across runtimes and script compatibility. For any version updates, document the changes in this appendix.

10.6 Step-by-step execution procedure

Describe how to run the simulations reproducibly:

- Preparation: synchronize system clocks; verify network connectivity; configure DDS_DOMAIN_ID=0 and the ports for each script.
- Launching Node 1: Run Launch_N1.py; enter the log file name and the publishing period (Ts_comm). Three consoles will open for N1PUB.py, N1SUB.py, and N1_csv.py.
- Launching Node 2: Run launch_N2.py; this will open the consoles for N2PUB.py and N2SUB.py.
- Simulink models: Load the corresponding models (S1, S2 or S_{ref}) and ensure that the base time (Ts_core) is 0.5 ms. Start the simulation once the scripts are running.
- Verification: Monitor the consoles to check for lost packets or "drops". Ensure that CSV files are being generated in the logs/ folder.
- Closure: Stop the Simulink models first, then the Python scripts; close the consoles to prevent any processes from running.

10.7 Raw data and nomenclature

The results are stored as CSV files in the project's logs/ folder. Each line contains the system clock timestamp in nanoseconds (t_wall_ns) and the signals (I_demand, V_SM, P_SM, I_SM) with nine decimal places. The naming convention is:

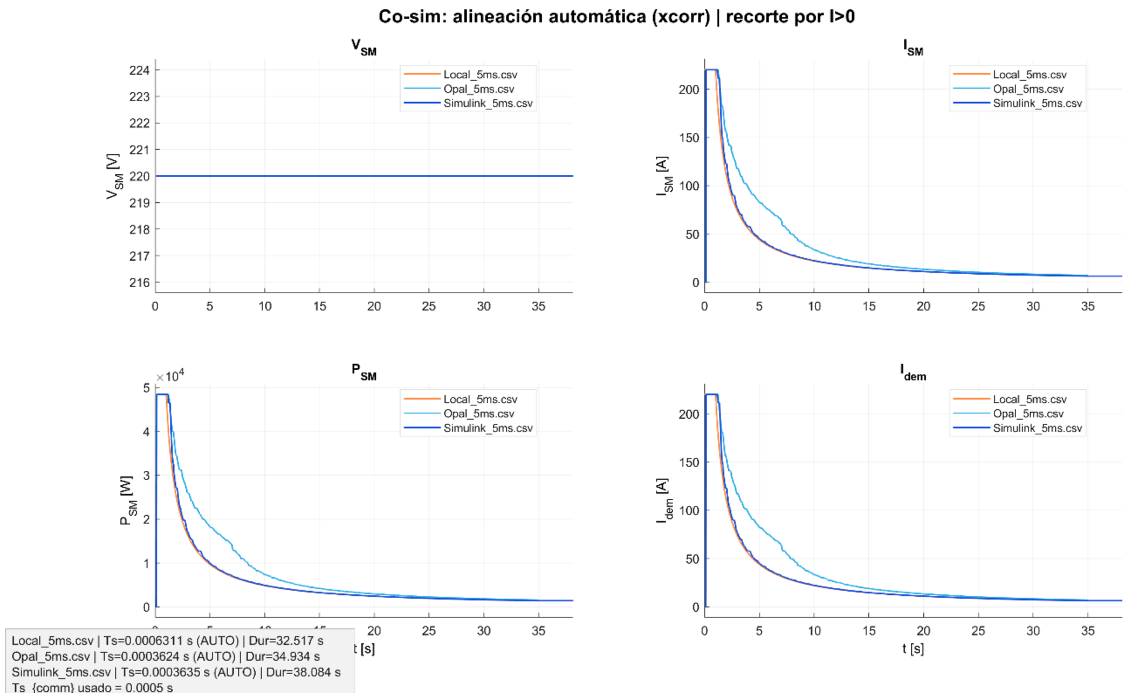
<base>_<Ts_comm>ms_<scenario>_<run>.csv

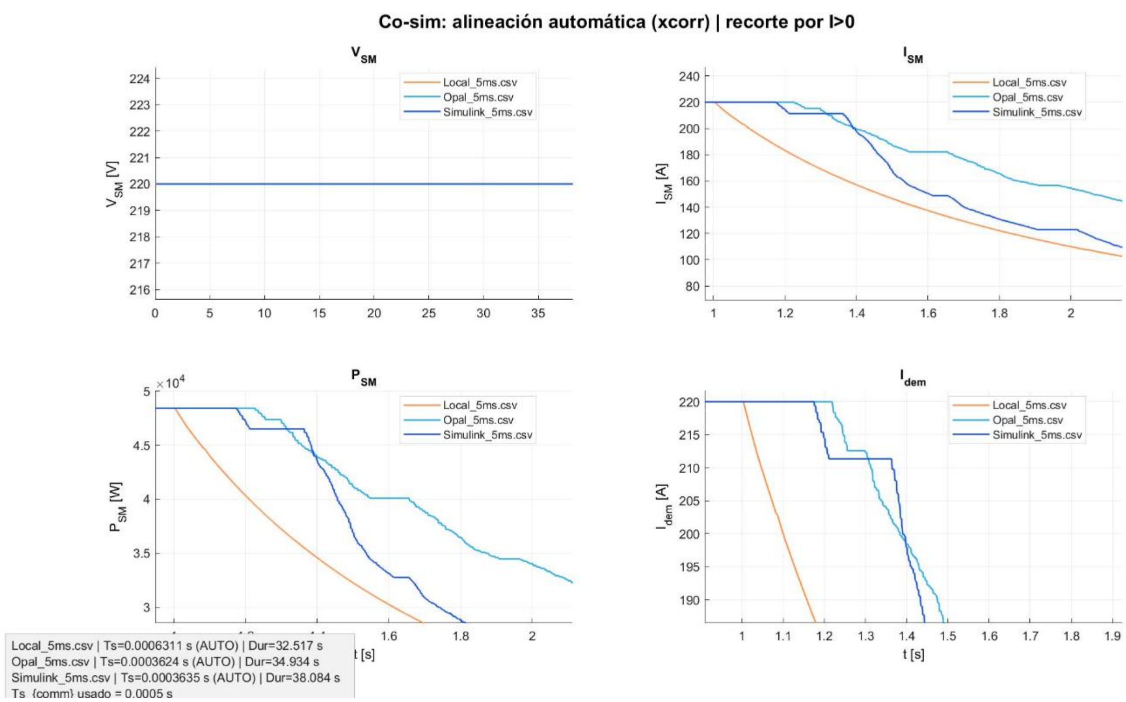
For example, exp1_0.5ms_S1_01.csv. A table will list the runs performed for each Ts_comm and scenario; it is not necessary to paste all the CSVs here. It is recommended to upload these files to a shared folder on Google Drive (e.g., drive:/CoSimulation/Resultados/CSV/), maintaining subfolders by scenario and date

10.8 High-resolution results figures

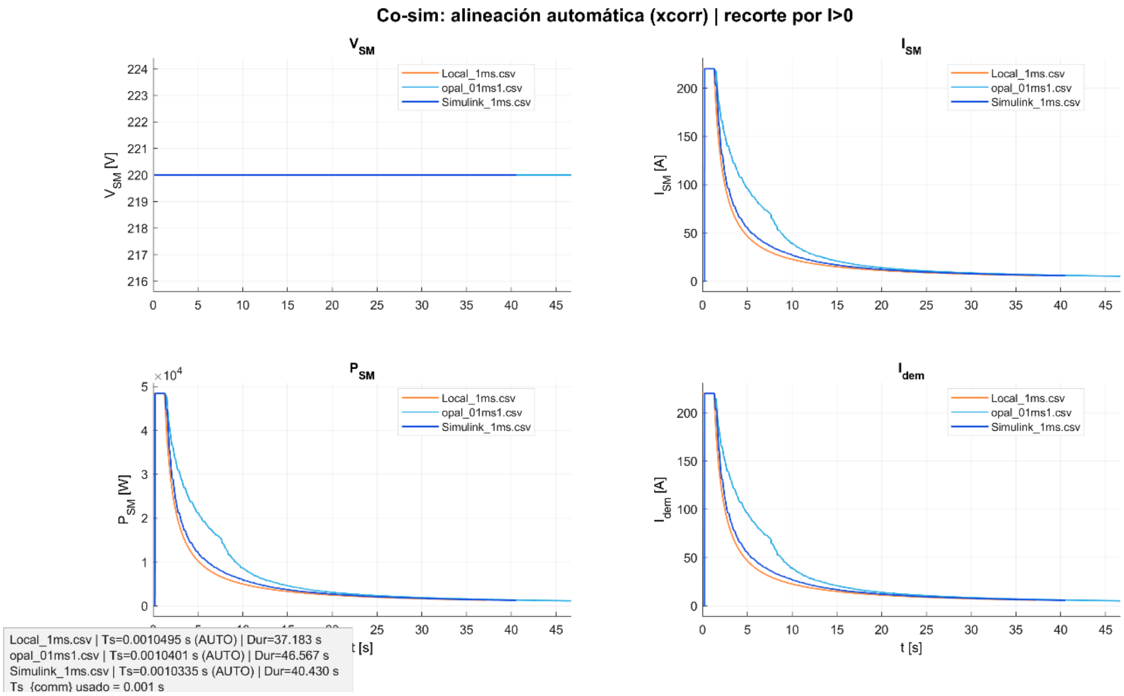
The results graphs are included in PNG/PDF format with standardized names (run_ts_comm_scenario_date.png). Each figure corresponds to a metric (e.g., latency vs. Ts_comm, active power vs. load, etc.). It is important that the figures have clear axes, titles, and legends.

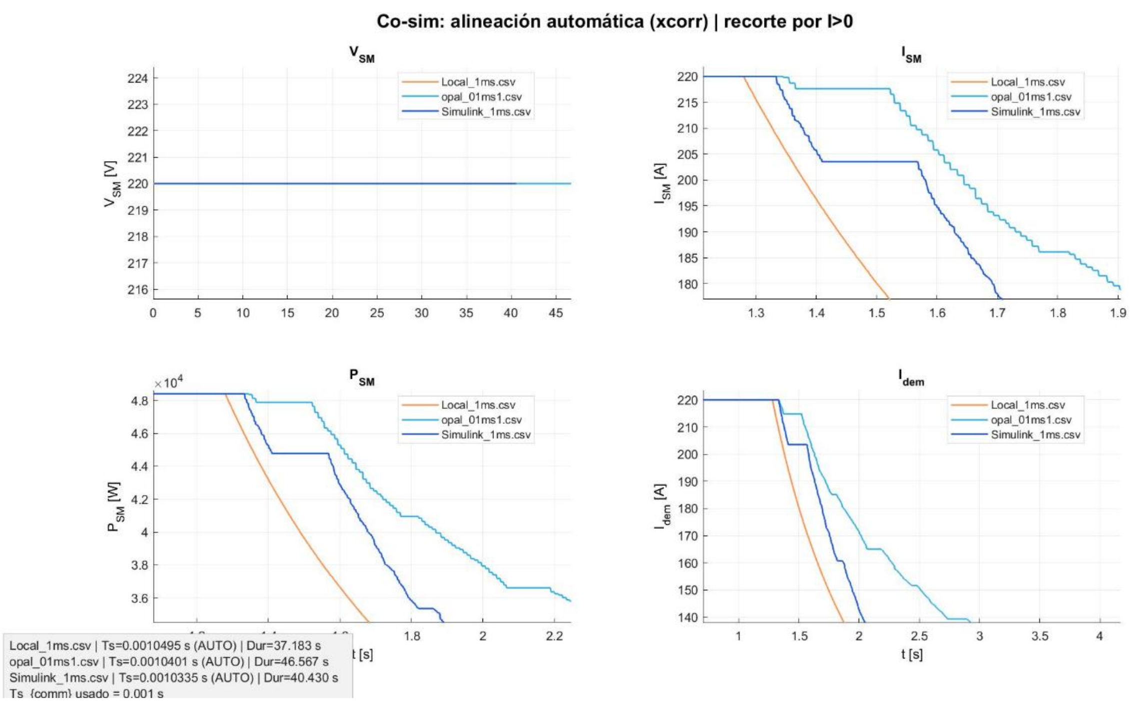
10.8.1 DC Tests Ts_comm = 0.0005



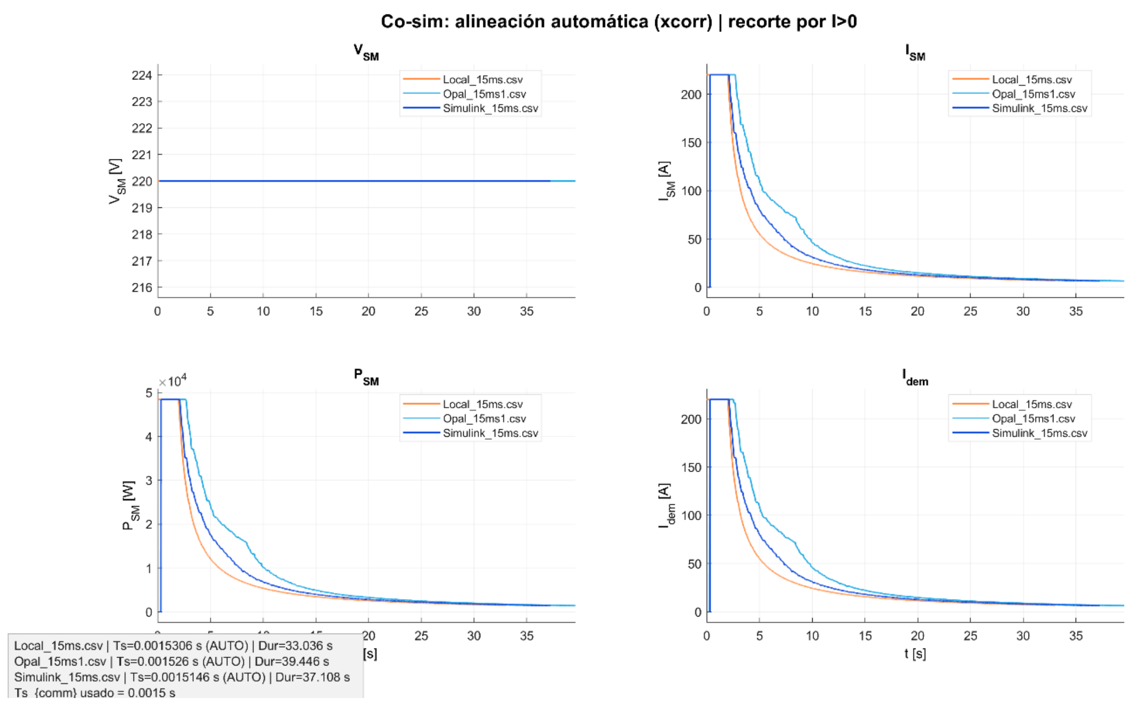


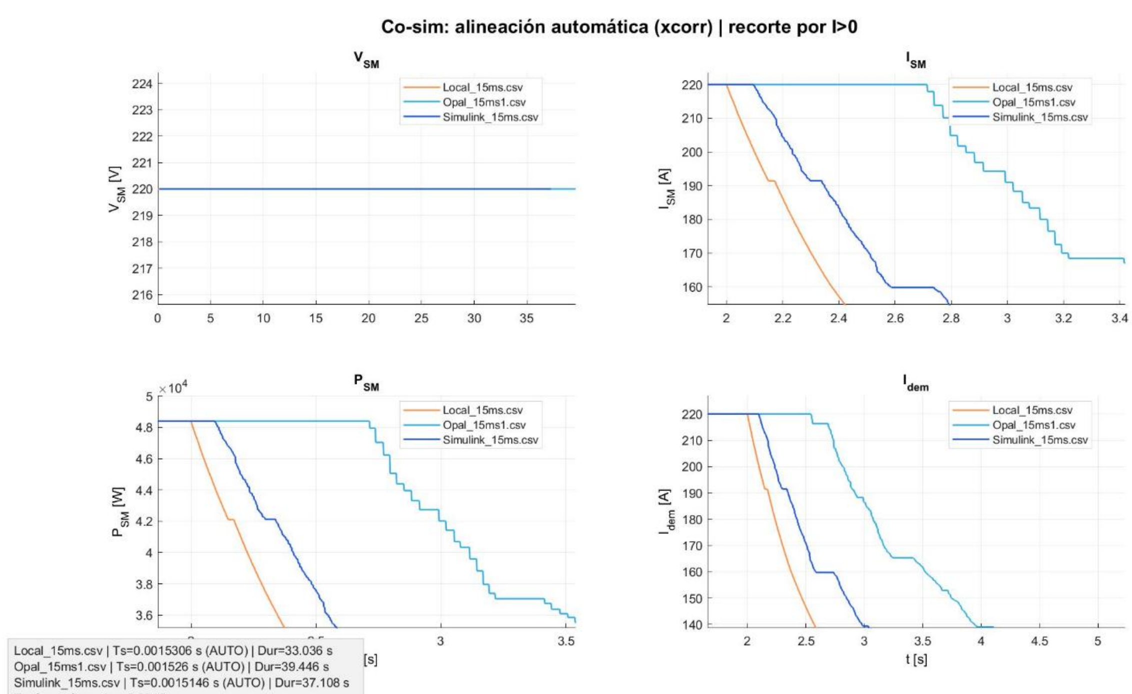
10.8.2 DC Tests Ts_comm = 0.001



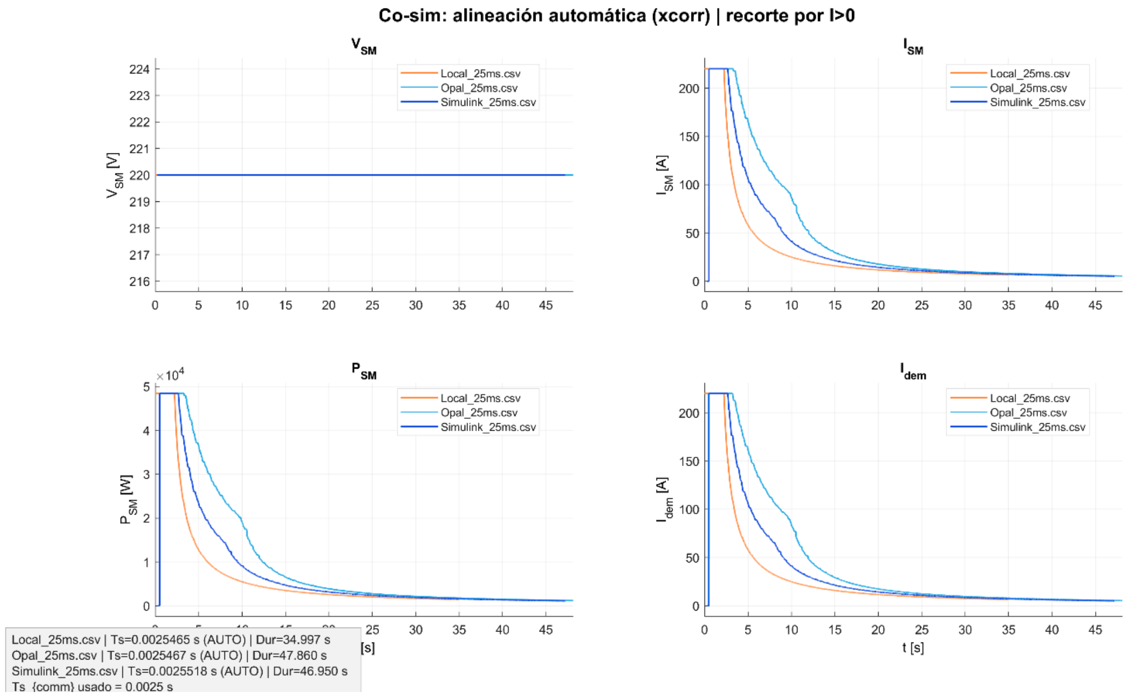


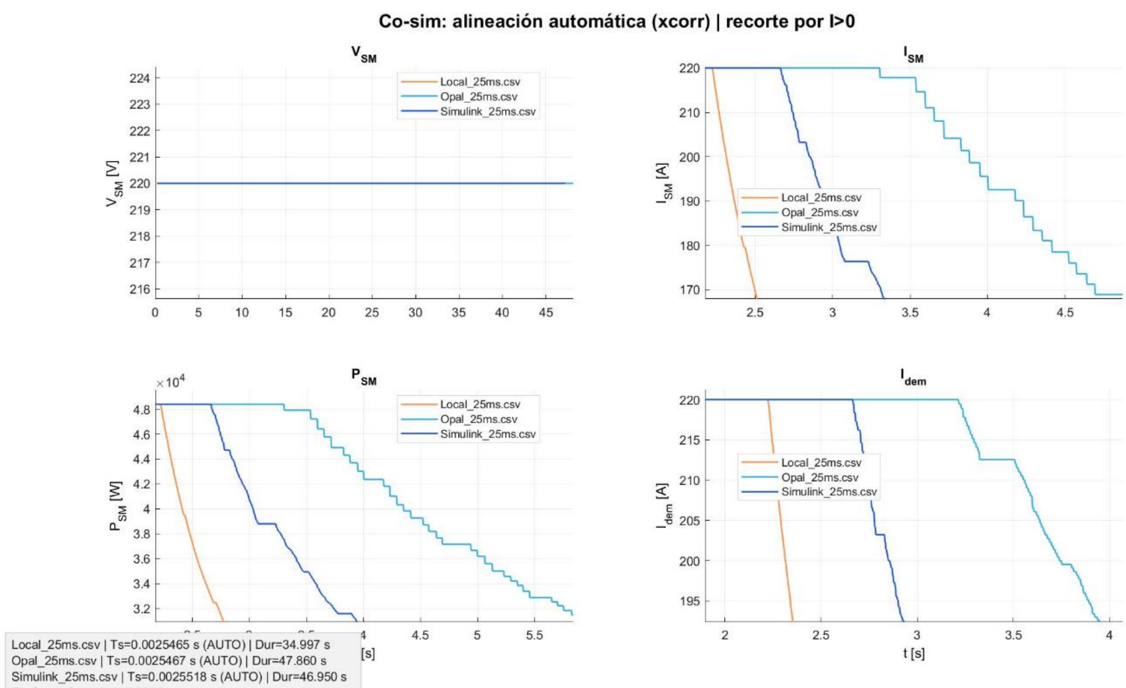
10.8.3 DC Tests Ts_comm = 0.0015



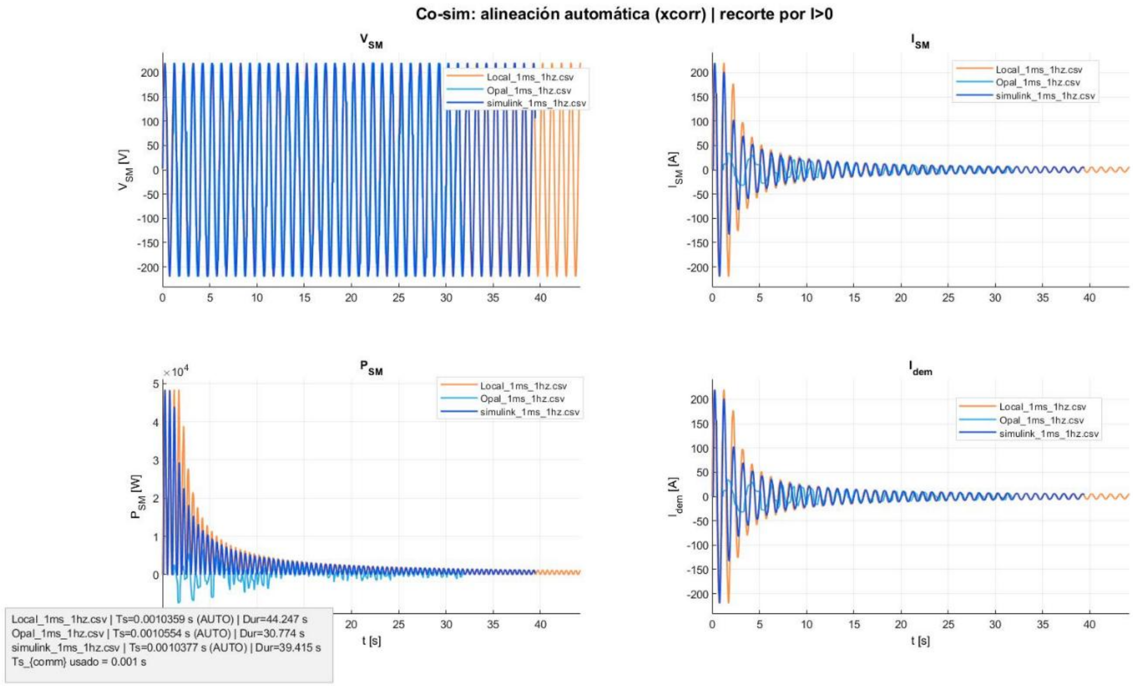


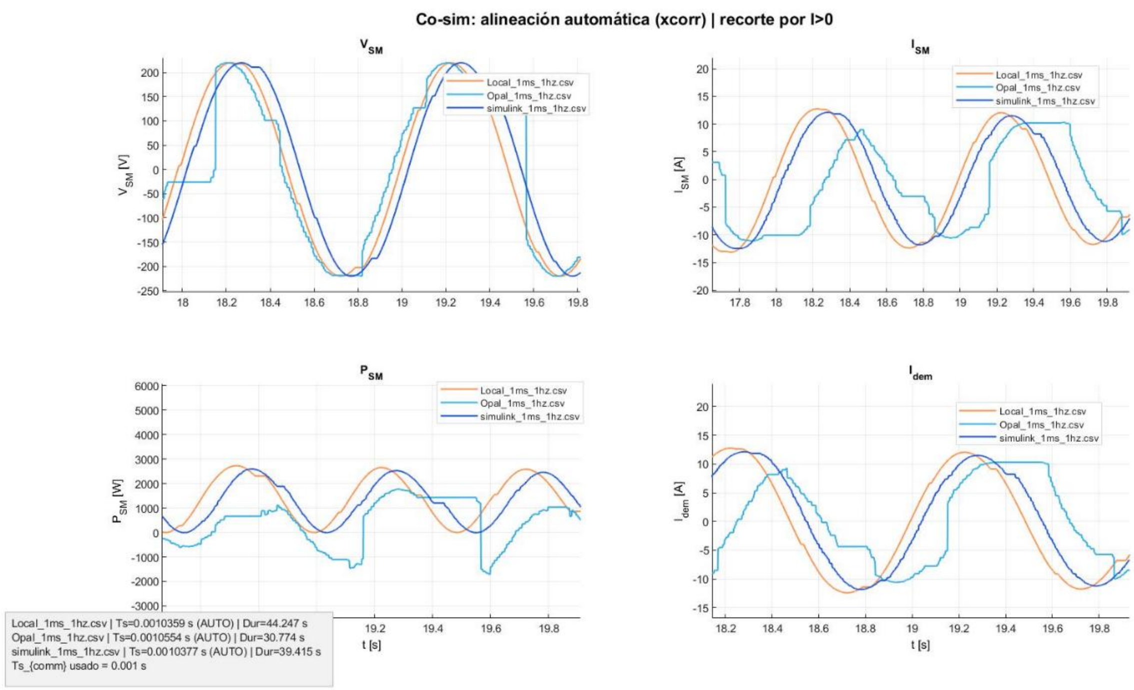
10.8.4 DC Tests Ts_comm = 0.0025



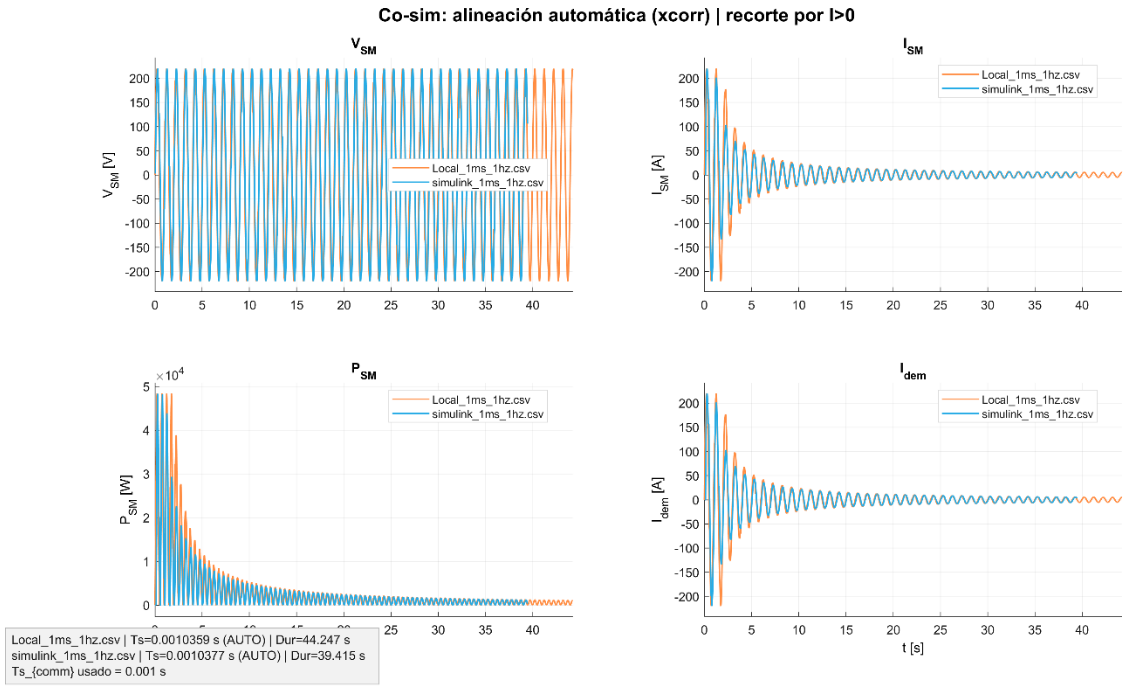


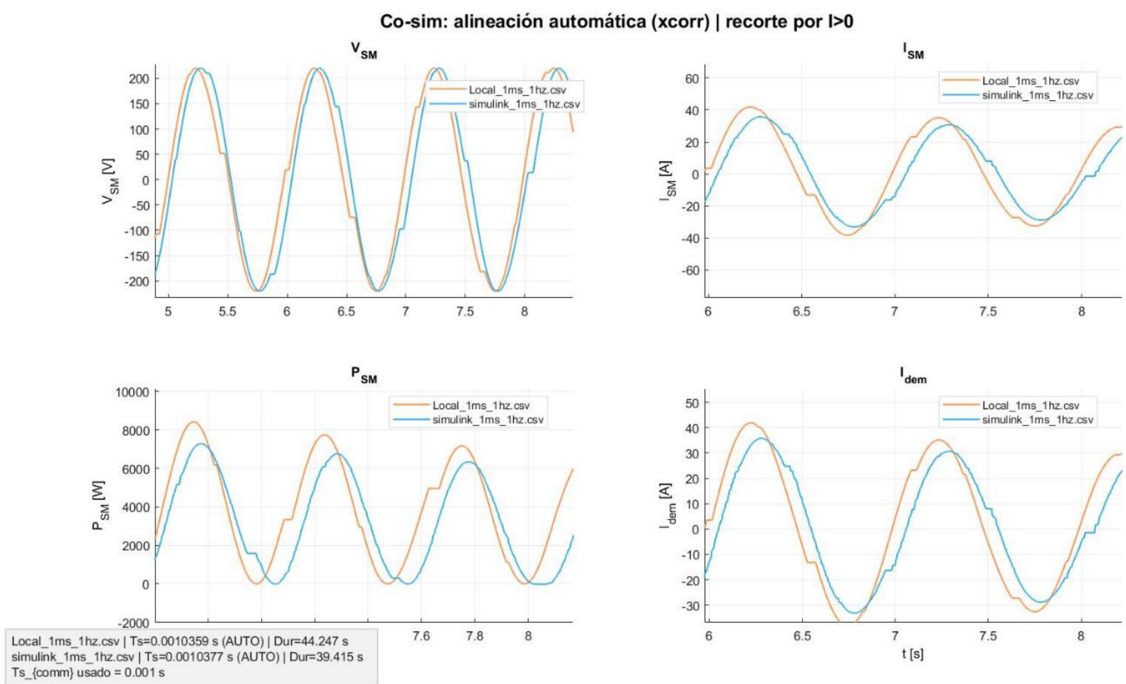
10.8.5 AC results for f = 1 Hz (Ts_comm = 1 ms, includes OPAL-RT)



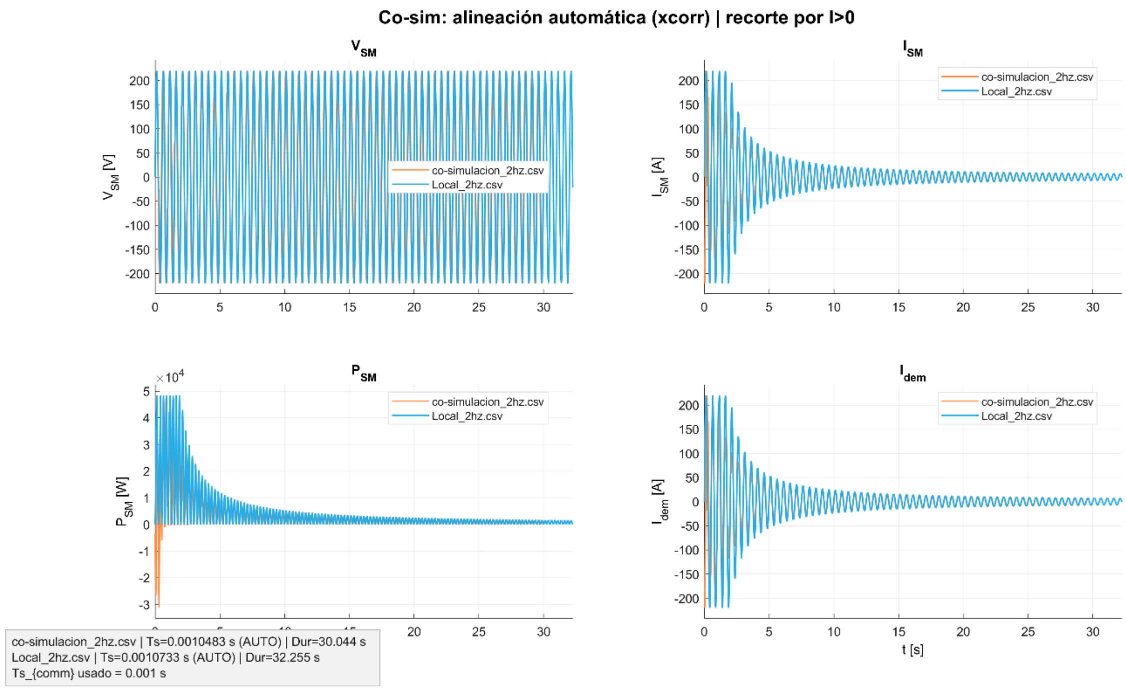


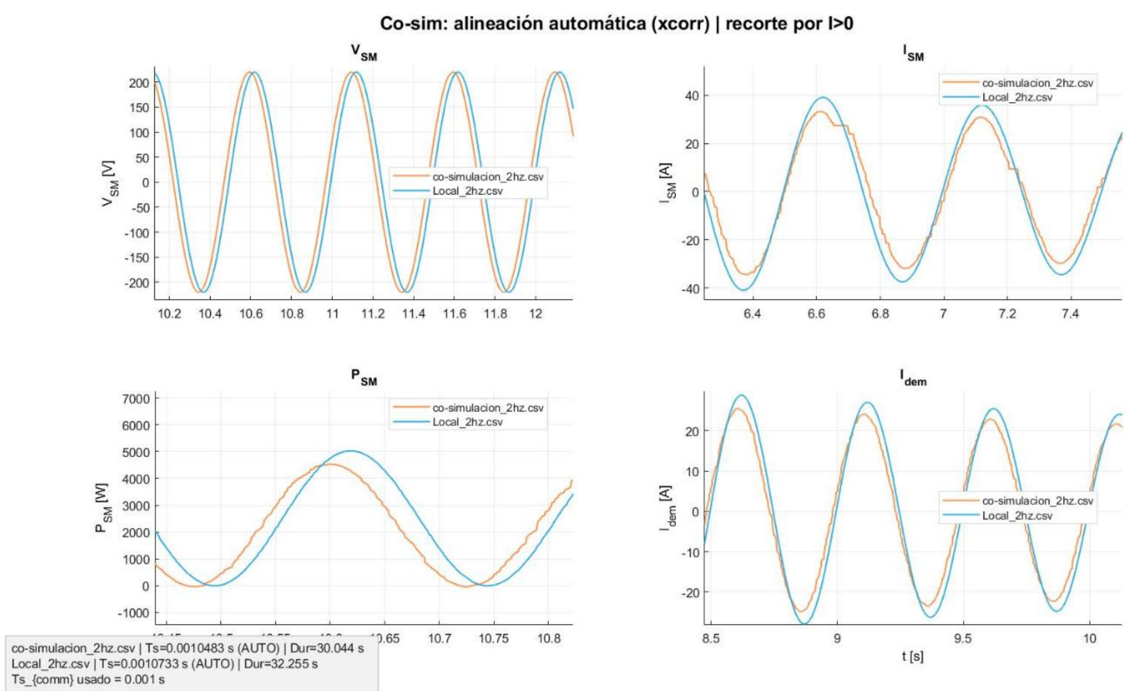
10.8.6 Results in AC for f = 1 Hz (Ts_comm = 1 ms)



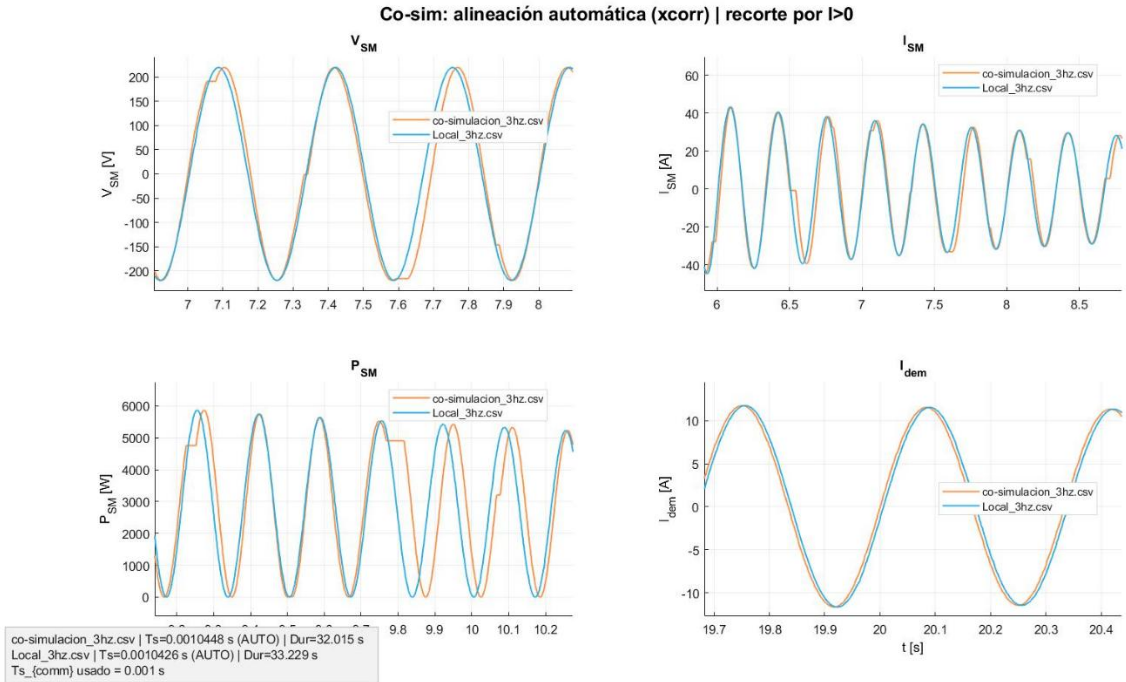


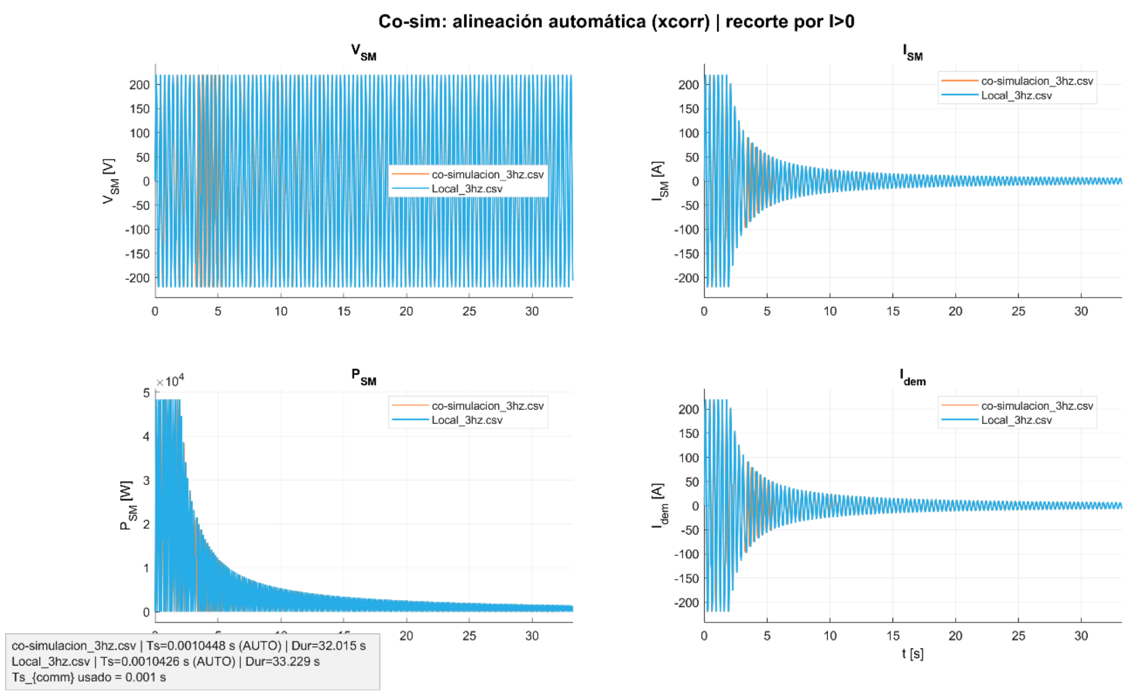
10.8.7 Results in AC for f = 2 Hz (Ts_comm = 1 ms)



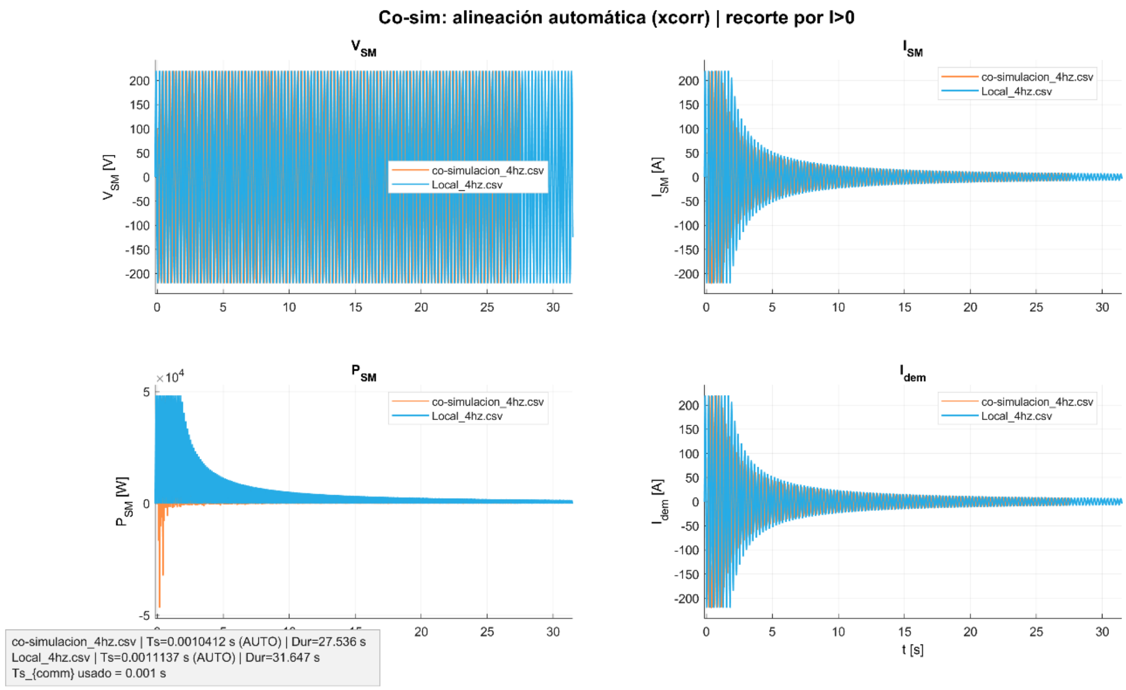


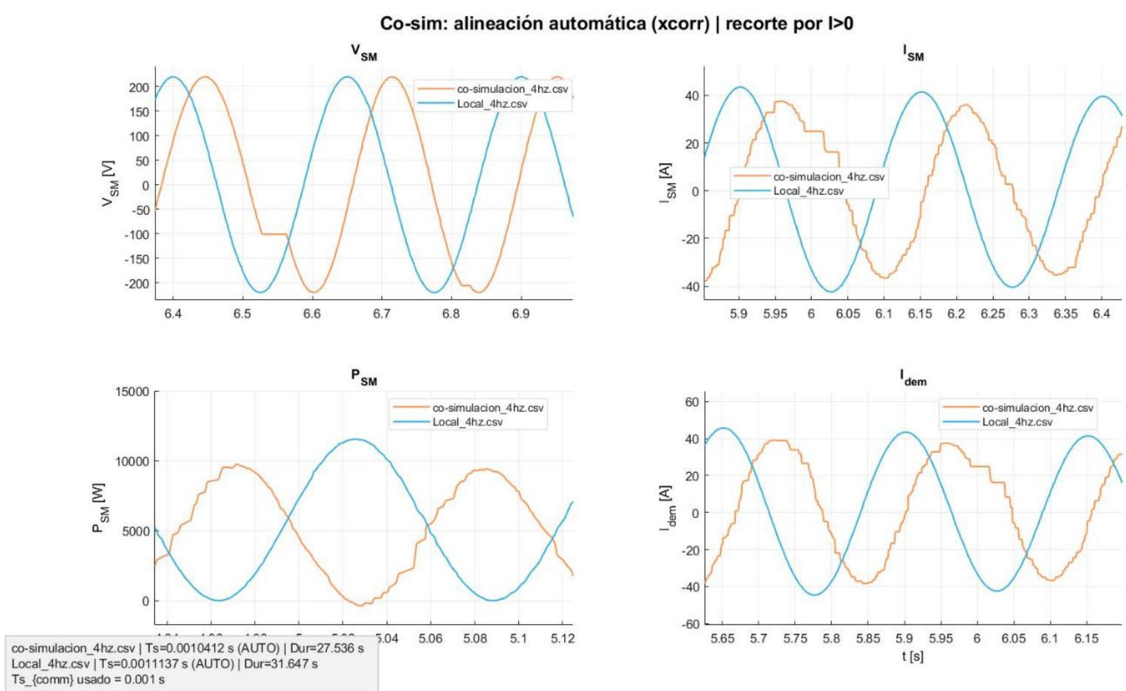
10.8.8 Results in AC for f = 3 Hz (Ts_comm = 1 ms)



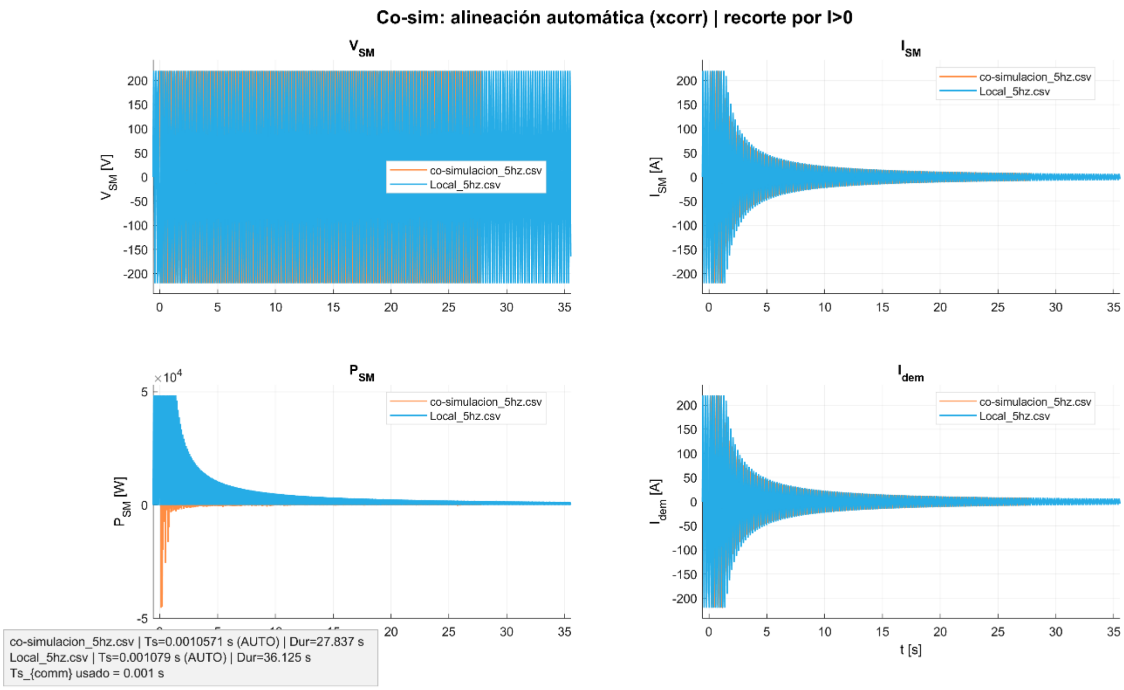


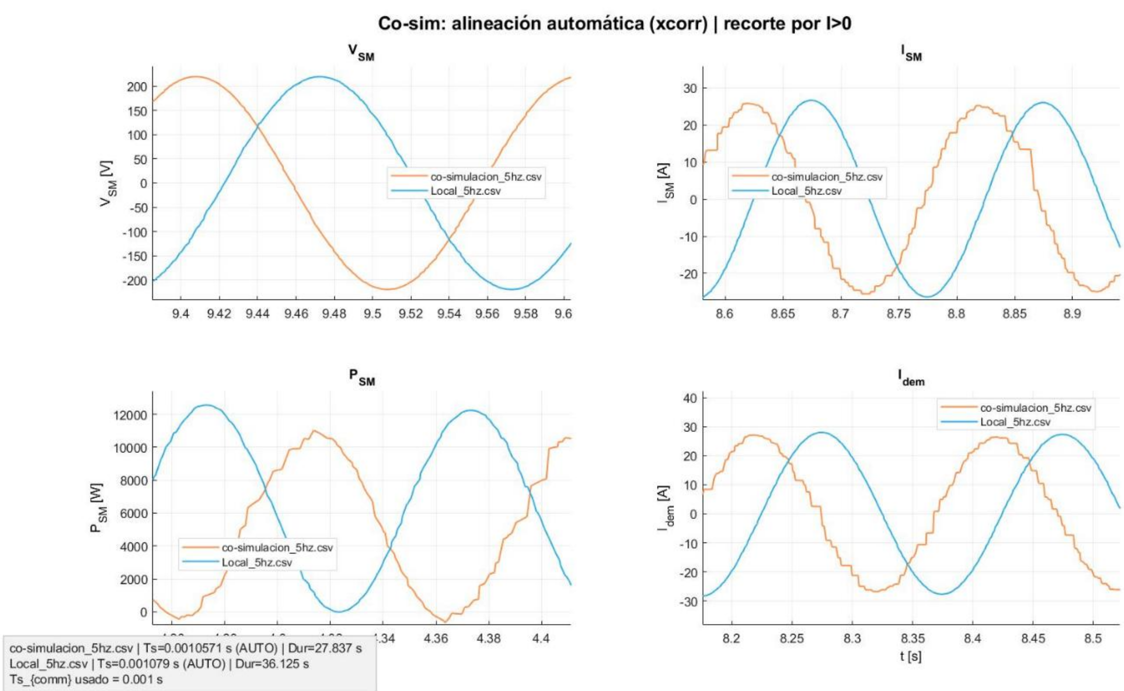
10.8.9 Results in AC for f = 4 Hz (Ts_comm = 1 ms)



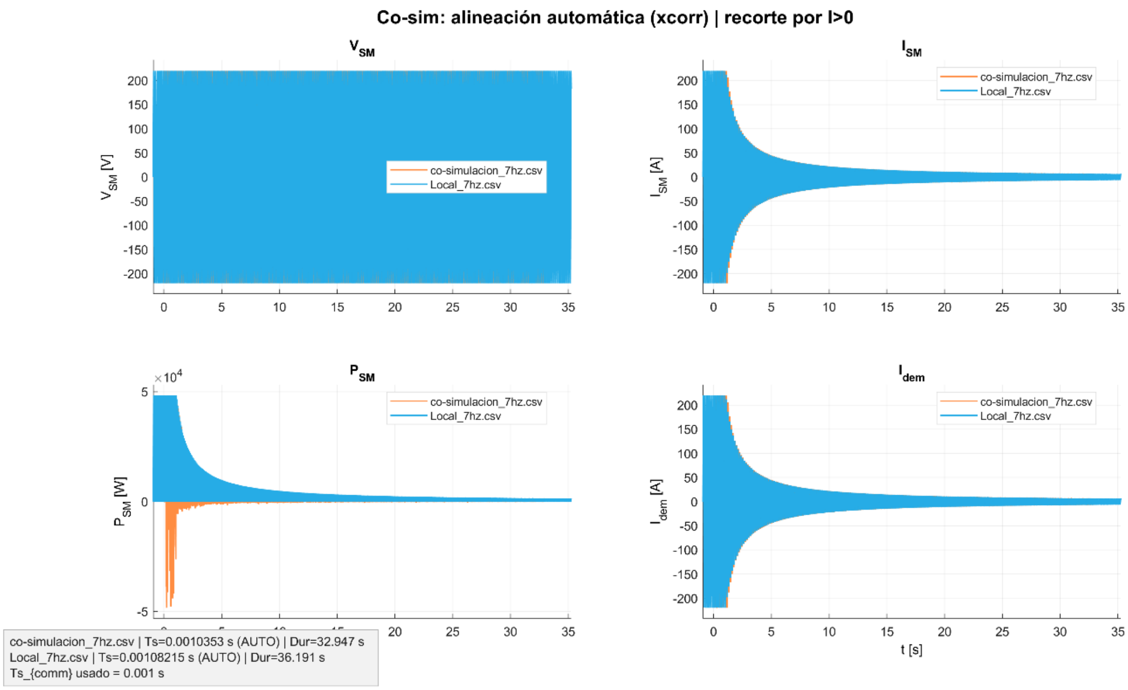


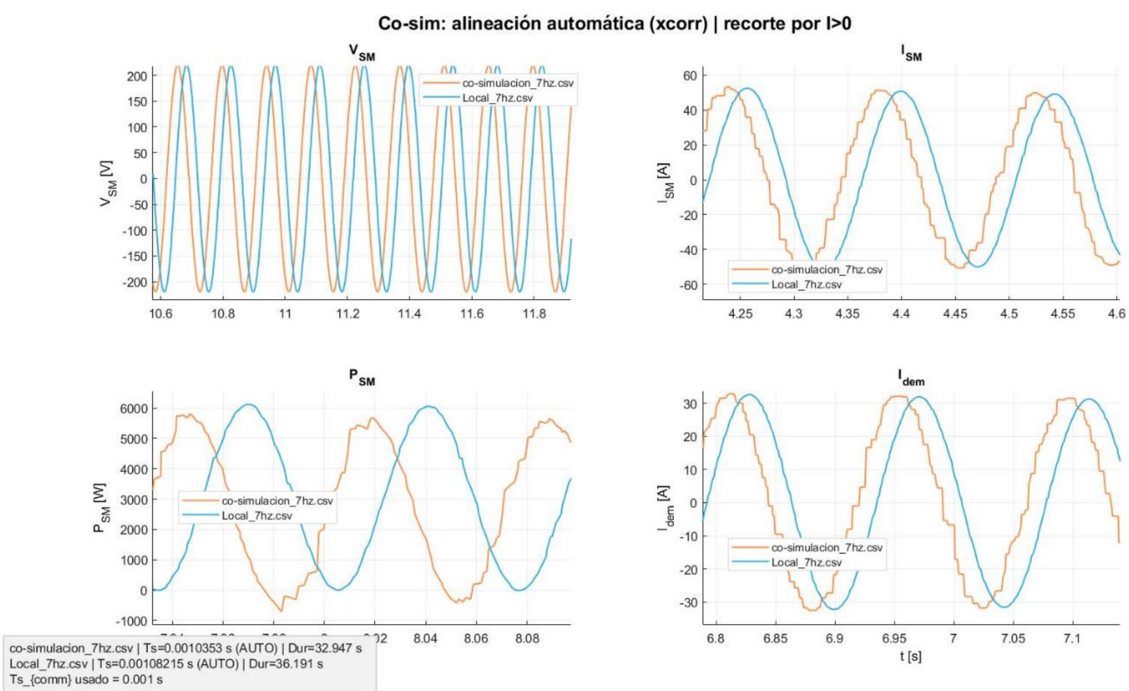
10.8.10 Results in AC for f = 5 Hz (Ts_comm = 1 ms)



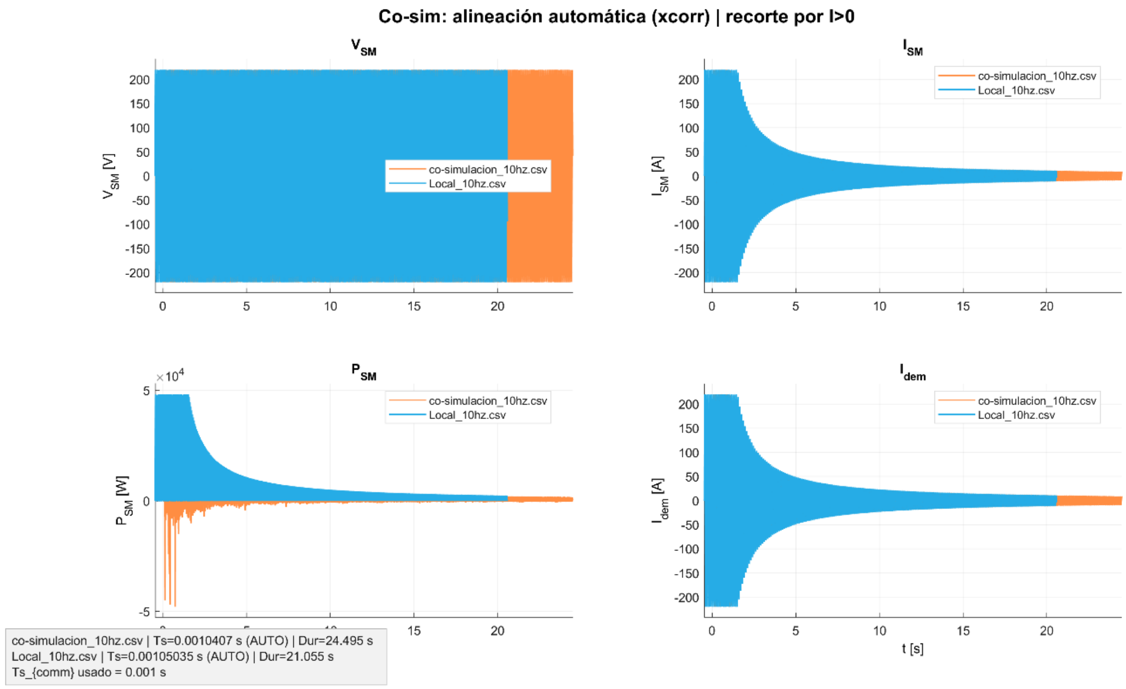


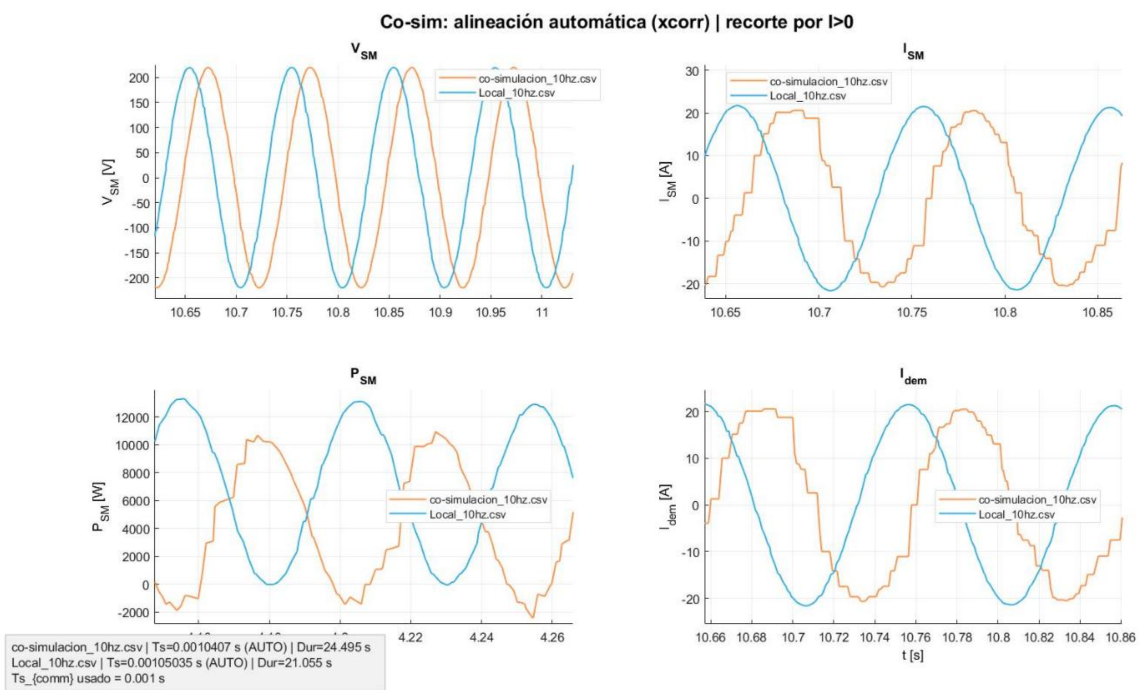
10.8.11 Results in AC for f = 7 Hz (Ts_comm = 1 ms)



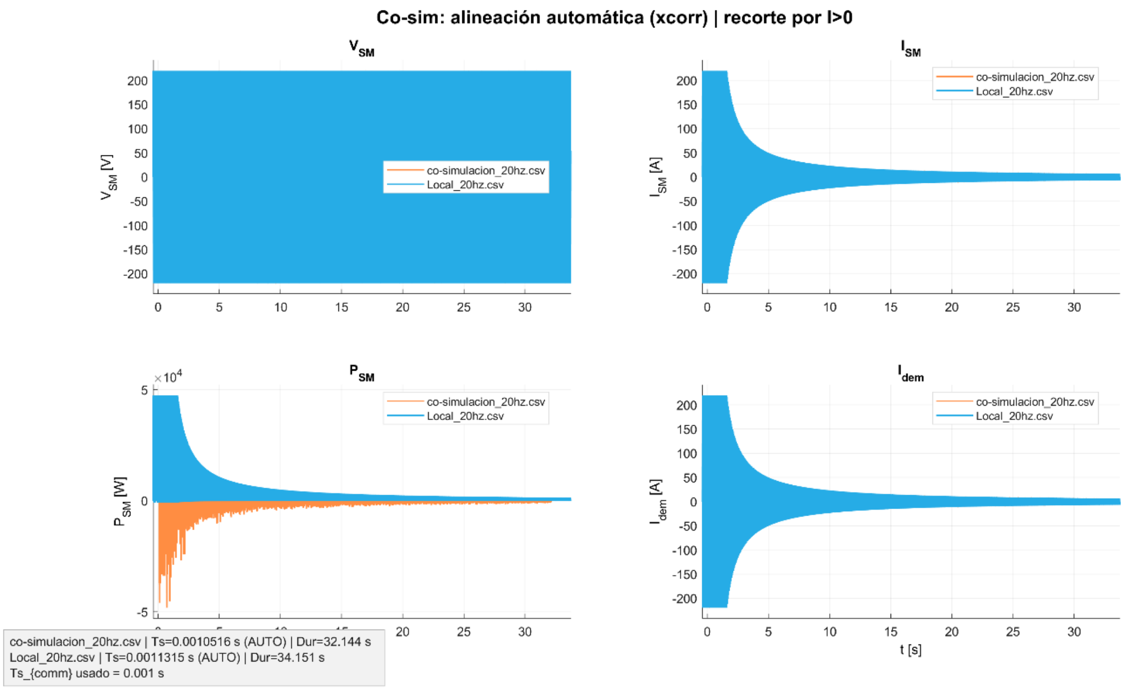


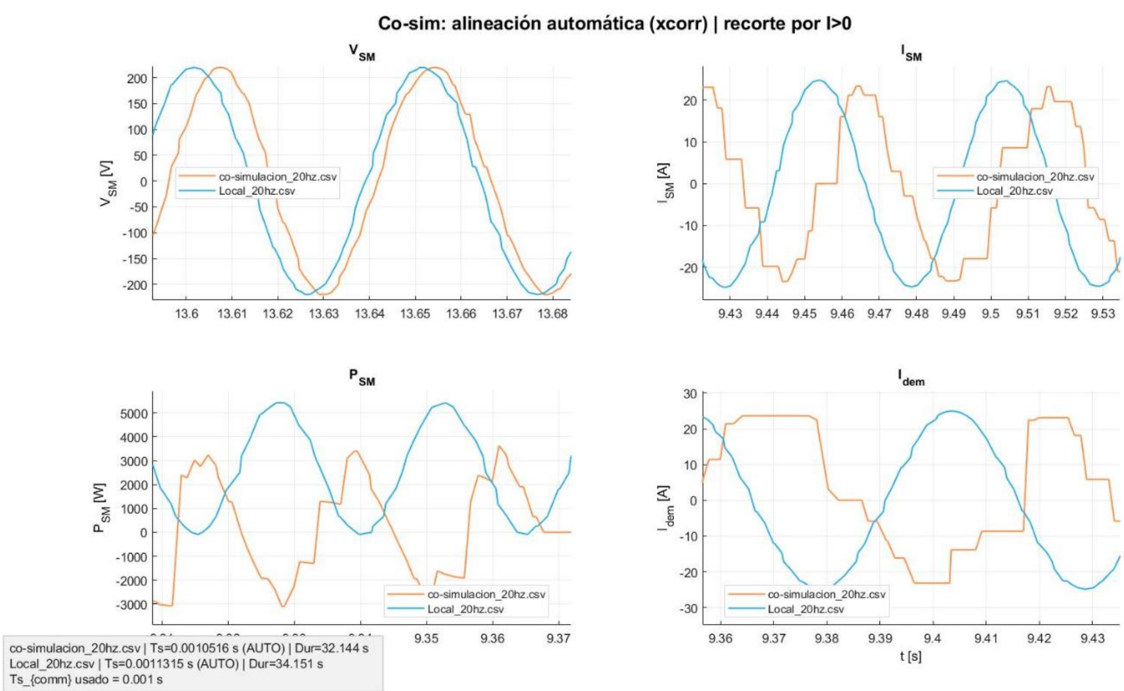
10.8.12 Results in AC for f = 10 Hz (Ts_comm = 1 ms)



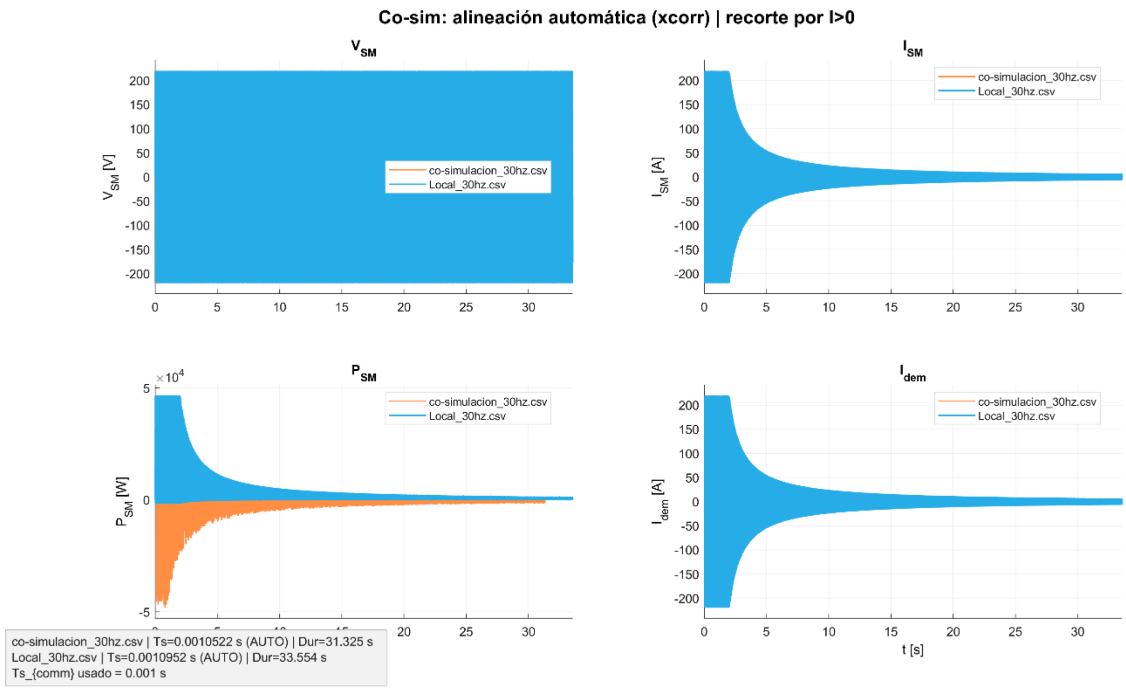


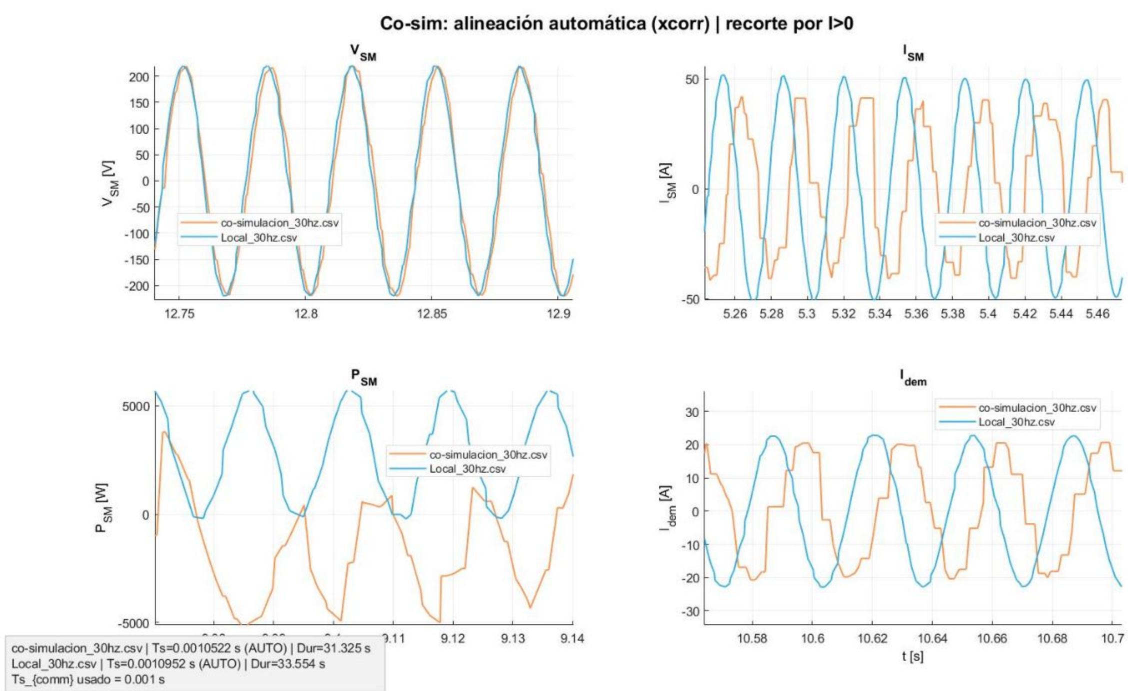
10.8.13 Results in AC for f = 20 Hz (Ts_comm = 1 ms)



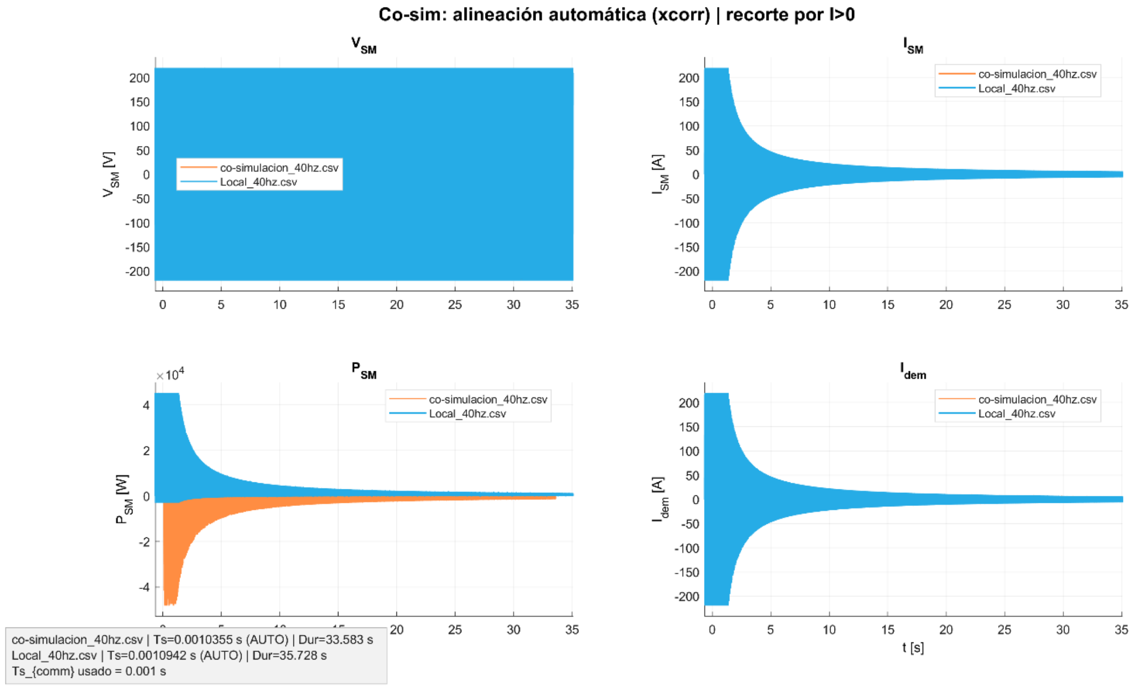


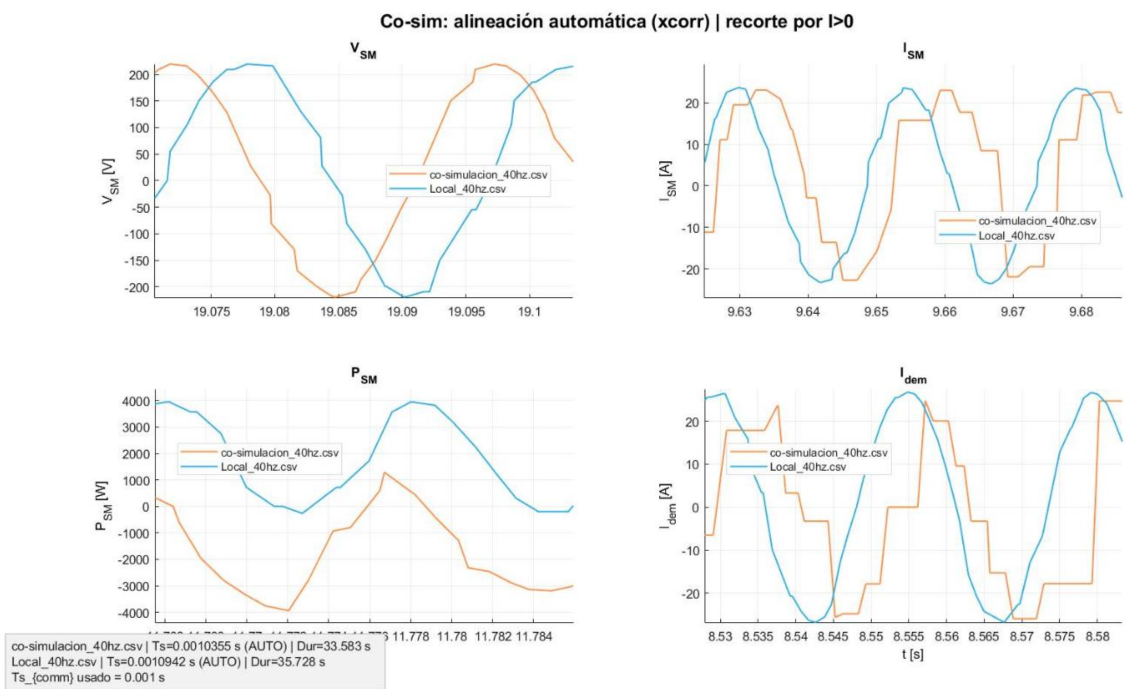
10.8.14 Results in AC for f = 30 Hz (Ts_comm = 1 ms)





10.8.15 Results in AC for f = 40 Hz (Ts_comm = 1 ms)





10.8.16 Results in AC for f = 50 Hz (Ts_comm = 1 ms)